

A Short Proof of the Quasi-Riemann Hypothesis

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Abstract

We prove that the Riemann zeta function, all Dirichlet L -functions, and all finite-order Hecke L -functions over $\mathbb{Q}(\sqrt{-3})$ are zero-free in $\operatorname{Re} s > 47/48$. Our proof combines cubic theta reflection, a quadratic large sieve, and Hecke zero-density estimates with Mellin continuation.

1 Introduction

The Quasi-Riemann Hypothesis asserts that there exists a constant $c > 0$ such that every nontrivial zero ρ of the Riemann zeta function satisfies $\operatorname{Re} \rho \leq 1 - c$. Here a nontrivial zero means a zero in $0 < \operatorname{Re} s < 1$. We give an argument with $c = 1/48$ that also treats Dirichlet L -functions and finite-order Hecke L -functions over $F = \mathbb{Q}(\sqrt{-3})$.

Theorem 1. *The Riemann zeta function has no zeros in $\operatorname{Re} s > 47/48$. More generally, every finite-order Hecke L -function over F , and every Dirichlet L -function, is zero-free in this half-plane, whether its character is primitive or imprimitive. For a principal character, the L -function has its usual simple pole at $s = 1$; this pole is allowed in the assertion.*

The boundary $\operatorname{Re} s = 47/48$ is not included in the zero-free assertion. Throughout, L -functions are uncompleted, with their usual meromorphic continuations; a zero is a point where the function is holomorphic and vanishes.

The supplied manuscript [1] states the stronger zero-free half-plane $\operatorname{Re} s > 7/8$. We adapt its probe and Mellin-continuation framework, in particular the constructions in its Sections 4, 6, and 7, to give a shorter argument for the weaker half-plane in Theorem 1. The inputs are the unconditional cubic theta formulas in [2, Section 5 and Appendix A], the quadratic Hecke large sieve in [3, Theorem 1.1 and Corollary 1.2], and the general Hecke zero-density estimate in [4, Lemma 3.3]. The conditional prime-sum and dispersion estimates in [2] are not used. Appendices A and B give the arithmetic and reflection calculations for the probe used here.

Poisson summation introduces finite-order twists over F , so we work with one zero supremum for this family. The norm pullback transfers the result to Dirichlet functions in Subsection 5.2.

Write $L_F(s, \eta)$ for the Hecke L -function of a character η over F , and let $\mathcal{P}_F$ be the set of primitive finite-order Hecke characters over F . Define the family zero supremum

$$\beta_* = \sup(\{1/2\} \cup \{\operatorname{Re} \rho : \eta \in \mathcal{P}_F, L_F(\rho, \eta) = 0, 1/2 \leq \operatorname{Re} \rho \leq 1\}).$$

This set is nonempty and contained in $[1/2, 1]$, so $1/2 \leq \beta_* \leq 1$. Absolute convergence of the Euler products also excludes zeros in $\operatorname{Re} s > 1$. Poles, including the principal pole at 1, are excluded from the definition. We do not assume that any zero or any character attains the supremum. Put

$$b = \frac{47}{48}, \quad C(r) = r - \frac{2}{3}, \quad m_0 = \frac{1}{1200}, \quad \delta_0 = \frac{m_0}{4} = \frac{1}{4800}.$$

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1.1 Proof overview

Suppose, for a contradiction, that $\beta_* > b$, and fix a primitive finite-order target character η over F , including the principal character. Section 2 constructs a normalized probe J_η , a principal signal f_η , and a holomorphic correction H_η . Their defining formulas and normalization appear together in Subsection 2.5. The proof has three ingredients:

1. **A small probe:** $J_\eta(Z) \ll_{\eta, \varepsilon} Z^{1/4+\varepsilon}$ for every $\varepsilon > 0$ and $Z \geq 2$ (Section 3).
2. **An accurate comparison:** $J_\eta(Z) - f_\eta(Z) \ll_\eta Z^{\beta_*-2/3-\delta_0}$ for $Z \geq 2$ (Section 4).
3. **A nonvanishing numerator:** $|H_\eta(s) - 1| \leq 1/2$ on $\operatorname{Re} s > 11/12$ (Subsection 2.5).

The saving δ_0 is independent of the target; the implied constants and the finite set of excluded primes may depend on it.

The first two estimates bound the signal; its decay at zero then continues its Mellin transform $e^{(s-5/6)^2} H_\eta(s)/L_F^S(s, \eta)$ to $\operatorname{Re} s > \beta_* - \delta_0$. The third permits division by H_η . The same positive improvement for every target contradicts the supremum. Section 5 supplies the analytic details.

1.2 Notation

Use the complex embedding specified by $\omega = e^{2\pi i/3}$, and write $\mathcal{O} = \mathbb{Z}[\omega]$ and $\lambda = 1 + 2\omega = \sqrt{-3}$. For $a \in \mathcal{O}$ put $q_a = N_{F/\mathbb{Q}}(a) = |a|^2$; explicitly, $q_{a_1+a_2\omega} = a_1^2 - a_1a_2 + a_2^2$. For a nonzero integral ideal $\mathfrak{a}$, put $q_{\mathfrak{a}} = N\mathfrak{a} = |\mathcal{O}/\mathfrak{a}|$. In particular, $q_{(a)} = q_a$ for $a \neq 0$. The ring is norm-Euclidean and hence every nonzero ideal is principal. Every nonzero ideal prime to $3\mathcal{O}$ has a unique generator congruent to 1 modulo $3\mathcal{O}$, which we call primary: reduction identifies the six units with $(\mathcal{O}/3\mathcal{O})^\times$. The unit group is $\mu_6 = \{\pm 1, \pm\omega, \pm\omega^2\}$. Lattice counting gives $O(X)$ elements of norm at most X for $X \geq 1$, and therefore $O(X)$ nonzero integral ideals of norm at most X .

A finite-order Hecke character modulo a nonzero integral ideal $\mathfrak{f}$ means a character of the ray class group modulo $\mathfrak{f}$; its continuous infinite type is trivial because $\mathbb{C}^\times$ is connected. Extend its ideal values by zero to ideals not prime to $\mathfrak{f}$, and use

$$L_F(s, \eta) = \sum_{\mathfrak{a} \neq 0} \eta(\mathfrak{a})(N\mathfrak{a})^{-s} = \prod_{\mathfrak{p}} (1 - \eta(\mathfrak{p})(N\mathfrak{p})^{-s})^{-1} \quad (\operatorname{Re} s > 1).$$

For an imprimitive presentation, this includes its prescribed deleted Euler factors. A superscript S means further deletion of the Euler factors at primes in S .

For each fixed target, the excluded set S contains the primes above six, the target conductor primes, and a fixed small-prime cutoff specified below. A good prime lies outside S .

Unqualified sums over a, b, c, n, s are ideal sums represented by their unique primary generators and avoid S , unless expressly stated otherwise. The variables m and u range over elements, including their unit factors. The superscript sf means squarefree, and $(a, S) = 1$ means that no prime ideal in S divides (a) (or the ideal a , when a denotes an ideal). We use the Mellin transform $\widehat{W}(t) = \int_0^\infty W(r)r^{t-1} dr$; an integral over (σ) runs upward on $\operatorname{Re} t = \sigma$. For $Z > 0$, powers Z^s use the real logarithm. An annular smooth weight belongs to $C_c^\infty((0, \infty))$. Implied constants may depend on the fixed target, S , the weights, and displayed small parameters; uniformity in moving rows and scales will be stated where needed.

2 The probe and its principal signal

Fix a primitive finite-order target η . Choose a cutoff P_0 large enough for the uniform product bound in Subsection 2.5, and let S contain the primes above six, the conductor primes of η , and all primes of norm at most P_0 . Fix this finite set before defining the probe. The constructions in this section, and the estimates in Sections 3 and 4, use this fixed target.

The cubic Gauss coefficients connect the probe to theta reflection, while Poisson summation exposes the reciprocal target L -function. Annular weights and a Mellin Gaussian control both representations.

2.1 Arithmetic and Fourier conventions

For $a \neq 0$ set $\alpha(a) = a/|a|$, and for $z \in F$ set $e(z) = \exp(2\pi i \operatorname{Tr}_{F/\mathbb{Q}}(z/\lambda))$. For planar Fourier transforms, extend the trace by $\operatorname{Tr}(w) = w + \bar{w}$ for $w \in \mathbb{C}$. The covolume of $\mathcal{O} \subset \mathbb{C}$ is $\sqrt{3}/2$, and its trace-pairing matrix in the basis $1, \omega$ is

$$(\operatorname{Tr}(v_i v_j / \lambda))_{i,j=1}^2 = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}, \quad (v_1, v_2) = (1, \omega).$$

This integer matrix has determinant -1 , so the lattice is self-dual for the pairing $e(zy)$. The corresponding self-dual measure is $d\mu(z) = (2/\sqrt{3}) dx dy$, with $z = x + iy$; our transform is $\mathcal{F}_{\mathbb{C}} f(y) = \int_{\mathbb{C}} f(z) e(-zy) d\mu(z)$. In particular, an $\mathcal{O}$ fundamental parallelogram has measure one. In real coordinates the pairing is $\operatorname{Tr}(zy/\lambda) = 2(\operatorname{Re} z \Im y + \Im z \operatorname{Re} y)/\sqrt{3}$; its matrix has determinant $-4/3$, giving precisely this self-dual measure.

For a good prime $\mathfrak{p}$ one has $q_{\mathfrak{p}} \equiv 1 \pmod{6}$, and reduction embeds μ_6 into $(\mathcal{O}/\mathfrak{p})^\times$. If $a \notin \mathfrak{p}$, define $\chi_{\mathfrak{p}}(a) = (a/\mathfrak{p})_6$ to be the unique root in μ_6 whose reduction is $a^{(q_{\mathfrak{p}}-1)/6}$; set $\chi_{\mathfrak{p}}(a) = 0$ if $a \in \mathfrak{p}$. For an ideal $\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{k_{\mathfrak{p}}}$ prime to S , define $\chi_{\mathfrak{a}}(v) = \prod_{\mathfrak{p}|\mathfrak{a}} \chi_{\mathfrak{p}}(v)^{k_{\mathfrak{p}}}$. For every integer j , the notation $\chi_{\mathfrak{a}}(v)^j$ means the usual power when v is a unit modulo $\mathfrak{a}$, and means zero otherwise, including when $j = 0$ or $j < 0$. At the unit ideal the empty product is $\chi_1(v) = 1$ for every v .

We distinguish the two orientations of the symbol: $\chi_{\mathfrak{a}}(v)$ is a residue character in the element v , whereas, for $u \in \mathcal{O} \setminus \{0\}$, $\psi_u(\mathfrak{n}) = \chi_{\mathfrak{n}}(u)$ is the Kummer Hecke character in the ideal $\mathfrak{n}$, on ideals prime to S and (u) . Our Artin convention uses arithmetic Frobenius: at a prime not dividing $6u$, its action on $u^{1/6}$ has ratio $(u/\mathfrak{p})_6$. This specifies the character of $F(u^{1/6})/F$, even when its order is a proper divisor of six. We retain the prescribed value zero on ideals meeting (u) , even if the primitive inducing character is unramified at such a prime. In expressions such as $\chi_R(n)$, the argument is the chosen primary generator of the ideal n .

2.2 Definition of the probe

For a nonzero modulus c , a residue character ψ modulo c extended by zero on nonunits, and $k \in \mathcal{O}$, put

$$g_{\psi}(c, k) = \sum_{d \bmod c} \psi(d) e(kd/c), \quad \gamma_j(c) = q_c^{-1/2} g_{\chi_c^j}(c, 1) \quad (c \text{ squarefree}).$$

In the definition of γ_j , we take $j \in \mathbb{Z}$ and $(c, S) = 1$. At modulus one these functions equal one. The additive character $d \mapsto e(d/c)$ is primitive: by self-duality, $e(hd/c) = 1$ for every $d \in \mathcal{O}$ holds exactly when $h \in c\mathcal{O}$. For squarefree c prime to S , each local cubic character $\chi_{\mathfrak{p}}^2$ is nonprincipal. The prime Gauss-sum identity $|g_{\chi_{\mathfrak{p}}^2}(p, 1)| = q_p^{1/2}$ and CRT therefore give $|\gamma_2(c)| = 1$, including $c = 1$.

A finite phase correction. For odd $c = a + b\omega$ (meaning $(c, 2) = 1$), define

$$\Gamma_q(c) = \frac{1 + i^{-b} + i^a + i^{b-a}}{2}, \quad G(c) = \overline{\chi_c(4)}\Gamma_q(c), \quad R(v, w) = \frac{\Gamma_q(vw)}{\Gamma_q(v)\Gamma_q(w)}. \quad (1)$$

Here $G(c)$ is used only for primary c prime to six, and $R(v, w)$ is defined for all odd v, w . Appendix A proves that $|G| = 1$ and that R is a symmetric sign-valued bicharacter. On primary arguments they descend to the ray class group $\mathcal{T}$ modulo $36\mathcal{O}$. The formula for R applies also to noncoprime pairs.

The physical probe. Let b_* be any fixed generator of $\prod_{\mathfrak{p} \in S} \mathfrak{p}$. By CRT choose a residue character ξ modulo b_* , extended by zero on nonunits, nonprincipal at every prime, and of order dividing six: use a quadratic character at odd residue characteristic and a cubic character at the prime of norm four. Define

$$\tau = q_{b_*}^{-1/2} g_\xi(b_*, 1), \quad \Xi(A) = \xi(A)\chi_A(b_*), \quad |\tau| = 1.$$

Here Ξ is a fixed finite ray character supported on S . To see this, evaluate ξ on primary generators and use the Kummer character $A \mapsto \chi_A(b_*)$; their conductors involve only primes in S . The local choices make ξ primitive modulo the squarefree modulus b_* , so the primitive Gauss-sum formula gives $|\tau| = 1$. For $A = cn^3$ set

$$T_{m,s,\eta}(t) = \sum_{c \text{ sf}, n} \gamma_2(c) \overline{\alpha(A)} \overline{G(A)} \eta(A) \Xi(A)^{-1} R(A, s) \chi_A(m) q_c^{-1/2-t} q_n^{-1-3t}.$$

Choose nonnegative nonzero smooth annular W_0, W_1 and put $\Phi(t) = e^{t^2}$. Extend the weights by zero at zero. For $X, Y, Z > 0$ define the probe by

$$I_\eta(X, Y, Z) = \frac{1}{Y} \sum_s \frac{W_1(q_s/Y) \chi_s(b_*)}{\tau \xi(s) \sqrt{q_{b_*} q_s X}} \sum_{m \in \mathcal{O}} \xi(m) q_s^{-1/2} g_{\chi_s}(s, -m) W_0\left(\frac{q_m}{q_{b_*} q_s X}\right) \times \frac{1}{2\pi i} \int_{(4)} Z^t \Phi(t) T_{m,s,\eta}(t) dt. \quad (2)$$

All inverse character values are evaluated only on elements prime to S . This expression is absolutely convergent. The annular weights leave only finitely many s and m , with $m \neq 0$. On $\text{Re } t = 4$, $|T_{m,s,\eta}(t)| \leq \sum_{c,n} q_c^{-9/2} q_n^{-13} < \infty$, uniformly in $\Im t$, and $|\Phi(4 + iv)| = e^{16-v^2}$. Thus the defining sums and integral may be interchanged.

2.3 Exact representations

Completed rows. Put

$$V_G(x) = \frac{e^{-(\log x)^2/4}}{2\sqrt{\pi}}, \quad \widehat{V}_G(t) = e^{t^2}.$$

For $V = V_G$ and a fixed finite-order ray character ν supported on S , define

$$C_V(Z; m, \nu) = \sum_{\substack{c \text{ squarefree}, b \text{ primary} \\ (cb, S)=1}} \frac{\gamma_2(c) \overline{\alpha(cb^3)} \nu(cb^3) \chi_{cb^3}(m)}{q_c^{1/2} q_b} V\left(\frac{q_c q_b^3}{Z}\right). \quad (3)$$

The variables c, b may share primes. The Gaussian integral gives the Mellin formula; for $A > 1/2$, the bound $V_G(x) \ll_A x^{-A}$ majorizes the series by $Z^A \sum_{c,b} q_c^{-1/2-A} q_b^{-1-3A} < \infty$, locally uniformly in $Z > 0$.

Exact low separation. For a class $\sigma \in \mathcal{T}$, choose a primary representative s_σ prime to S . Such representatives exist by CRT, including the conditions of avoiding the finitely many additional primes in S . Use the finite Fourier expansion and character

$$\overline{G(A)} = \sum_{\theta \in \widehat{\mathcal{T}}} a_\theta \theta(A), \quad \nu_\sigma(A) = \eta(A) \Xi(A)^{-1} R(A, s_\sigma).$$

Here $a_\theta = |\mathcal{T}|^{-1} \sum_{\gamma \in \mathcal{T}} \overline{G(\gamma)} \theta(\gamma)$. The bicharacter property makes $A \mapsto R(A, s_\sigma)$ a character, independent of the chosen representative. Put

$$B_{m,\sigma}(Z) = \sum_{\theta} a_\theta C_{V_G}(Z; m, \nu_\sigma \theta).$$

Applying Mellin inversion to Φ term by term gives $\frac{1}{2\pi i} \int_{(4)} Z^t \Phi(t) T_{m,s,\eta}(t) dt = B_{m,\sigma}(Z)$ for $s \in \sigma$. The characters $\nu_\sigma \theta$ form a fixed finite family supported on S , so the completed-row estimate applies with every original zero mask. For $Q_0 = q_{b_*} XY$ define

$$A_{m,\sigma,v}(Y) = \frac{1}{Y} \sum_{s \in \sigma} \frac{W_1(q_s/Y) \chi_s(b_*)}{\tau \xi(s)} (q_s/Y)^{-1/2+iv} q_s^{-1/2} g_{\chi_s}(s, -m).$$

Choose $\Omega \in C_c^\infty((0, \infty))$ equal to one on $\text{supp}(W_0) \text{supp}(W_1)$, the set of products of points in the two supports. On the original support, $q_m/Q_0 = (q_m/(q_{b_*} q_s X))(q_s/Y)$ lies in this fixed compact set. Mellin inversion gives the exact identity

$$I_\eta = \frac{Q_0^{-1/2}}{2\pi} \sum_{\sigma} \int_{\mathbb{R}} \widehat{W}_0(iv) \sum_{m \neq 0} \Omega(q_m/Q_0) \xi(m) (q_m/Q_0)^{-iv} A_{m,\sigma,v}(Y) B_{m,\sigma}(Z) dv. \quad (4)$$

Indeed, $q_m/(q_{b_*} q_s X) = (q_m/Q_0)/(q_s/Y)$ and $(q_{b_*} q_s X)^{-1/2} = Q_0^{-1/2} (q_s/Y)^{-1/2}$. Insert $W_0(r) = (2\pi)^{-1} \int_{\mathbb{R}} \widehat{W}_0(iv) r^{-iv} dv$. Rapid Mellin decay and the finite m support justify all interchanges; annular support removes $m = 0$.

Exact Poisson separation. Let $\widetilde{W}_0(|y|^2)$ be the planar Fourier transform of $W_0(|y|^2)$ and set

$$M(z) = \int_0^\infty \widetilde{W}_0(r) r^{z-1} dr, \quad \widehat{W}_1(w) = \int_0^\infty W_1(r) r^{w-1} dr.$$

The planar transform is radial and Schwartz. Consequently M is holomorphic on $\text{Re } z > 0$, and integration by parts gives rapid vertical decay on compact substrips of that half-plane. The transform $\widehat{W}_1$ is entire with the same vertical decay on every fixed strip. The same probe has the following representation, where u ranges over nonzero sixth-power-free elements prime to S , including all six units. On the lines $\text{Re } x = 3$, $\text{Re } w = 3$, and $\text{Re } z = 2$, one has

$$I_\eta = \frac{1}{(2\pi i)^3} \int_{(3)} \int_{(3)} \int_{(2)} X^{1/2-z} Z^{x+z-1} Y^{w-1} \Phi(x+z-1) M(z) \widehat{W}_1(w) \\ \times \sum_u^{(6)} q_u^{-z} \overline{\xi(u)} \mathcal{F}_{\eta,u}(x, w, z) dz dw dx. \quad (5)$$

Here $\mathcal{F}_{\eta,u}$ factors as

$$\mathcal{F}_{\eta,u}(x, w, z) = \frac{\zeta_F^S(6z) L_F^S(w, \psi_u)}{L_F^S(x, \eta \psi_u)} \mathcal{H}_{\eta,u}(x, w, z).$$

(6)

The Euler product $\mathcal{H}_{\eta,u}$ is given in Subsection A.3. Appendix A derives (5)–(6) from (2) and justifies absolute convergence on these initial lines. The row $u = 1$ has target denominator $L_F^S(x, \eta)$.

2.4 Row characters and their conductors

Choose a finite conjugation-stable family Θ of finite-order Hecke characters presented with moduli supported in S , containing $1, \eta, \bar{\eta}$. For a sixth-power-free element u prime to S , let $\mathcal{X}_u$ consist of the presentations

$$\psi_{u,\nu,\epsilon}(\mathfrak{n}) = \nu(\mathfrak{n})\psi_u(\mathfrak{n})^\epsilon, \quad \nu \in \Theta, \quad \epsilon \in \{1, -1\}.$$

All prescribed zero extensions are retained. If ψ^* is the primitive inducing character, then

$$L_{\text{orig}}(s, \psi) = L(s, \psi^*) \prod_{\mathfrak{p} \in E_\psi} (1 - \psi^*(\mathfrak{p})(N\mathfrak{p})^{-s}), \quad E_\psi \subset S \cup \{\mathfrak{p} : \mathfrak{p} \mid u\}.$$

Lemma 2 (Conductor and multiplicity). *For each presentation in $\mathcal{X}_u$ with $U \leq Nu < 2U$, the primitive conductor has norm $O_{S,\Theta}(U)$. For any fixed primitive character, there are at most twelve distinct elements u for which some member of $\mathcal{X}_u$ induces it. This counts physical rows, not presentations (u, ν, ϵ) . No member of a nonunit row $\mathcal{X}_u$ induces the principal character.*

Proof. At $\mathfrak{p} \notin S$, put $q = N\mathfrak{p}$ and let $\lambda_{\mathfrak{p}}(x) = (x/\mathfrak{p})_6$ on local units, using the lift from the residue field to μ_6 fixed above. This character has exact order six. Tame local reciprocity identifies the restriction of the Kummer character of u to local units with $\lambda_{\mathfrak{p}}^{\kappa v_{\mathfrak{p}}(u)}$, where the sign $\kappa \in \{1, -1\}$ depends only on the Artin convention. The unit part of u does not enter this restriction. Twists in Θ are unramified there. For orientation ϵ , the exponent is multiplied by ϵ . These characters are trivial on the principal units $1 + \mathfrak{p}$. Thus the conductor exponent is one precisely when $1 \leq v_{\mathfrak{p}}(u) \leq 5$, and zero otherwise.

The conductor at S is uniformly bounded: if $m > 2v_{\mathfrak{p}}(6)$, Hensel's lemma, applied to $X^6 - a$ at $X = 1$, shows that every $a \in 1 + \mathfrak{p}^m$ is a sixth power. Every local Kummer character is trivial there; the remaining twists have fixed conductors. The conductor therefore divides a fixed ideal times $\text{rad}(u)$, whose norm is at most Nu .

For a fixed inducing character and orientation ϵ , its inertia restrictions determine each valuation $v_{\mathfrak{p}}(u) \in \{0, \dots, 5\}$ uniquely. There are no S -prime factors. Hence they determine at most one ideal (u) , which has six generators. This conclusion is independent of $\nu \in \Theta$, since those twists have trivial inertia outside S ; there is no additional $|\Theta|$ factor in the count of distinct rows. The two orientations give the bound twelve. Nontrivial inertia at any prime of a nonunit proves the last assertion. $\square$

2.5 The principal row and the numerator

We identify the principal numerator row and prove that its correction H_η is nonvanishing on the half-plane needed for continuation.

Lemma 2 excludes nonunit rows with principal numerator. To exclude the five nonidentity units, take the element $\pi = -2 - 3\omega$, which is primary and has norm seven. By CRT choose

$$n \equiv \pi \pmod{36\mathcal{O}}, \quad n \equiv 1 \pmod{\mathfrak{p}} \quad (\mathfrak{p} \in S, \mathfrak{p} \nmid 6).$$

Then n is primary, prime to S , and $Nn \equiv 7 \pmod{36}$. For a unit $u \in \mu_6$, the defining symbol gives $\chi_{\mathfrak{p}}(u) = u^{(N\mathfrak{p}-1)/6}$. If $(n) = \prod \mathfrak{p}^{k_{\mathfrak{p}}}$, write $N\mathfrak{p} = 1 + 6h_{\mathfrak{p}}$. Reduction modulo 36 gives $Nn \equiv 1 + 6 \sum k_{\mathfrak{p}} h_{\mathfrak{p}}$, so $\chi_n(u) = u^{\sum k_{\mathfrak{p}} h_{\mathfrak{p}}} = u^{(Nn-1)/6} = u$. This calculation applies to composite n as well. Thus each of the five nonidentity units has a nonprincipal numerator, even after an arbitrary finite enlargement of S .

Thus $u = 1$ is the only principal numerator row. To prove positivity of $M(1/6)$, use, for $0 < \sigma < 1$,

$$|y|^{2\sigma-2} = \frac{1}{\Gamma(1-\sigma)} \int_0^\infty t^{-\sigma} e^{-t|y|^2} dt.$$

Pair this identity with $\widetilde{W}_0(|y|^2)$ using $d\mu(y)$. Absolute integrability follows near zero from $\sigma > 0$ and at infinity from Schwartz decay, and justifies the first interchange. Put $\kappa_F = 2\pi/\sqrt{3}$. With our Fourier convention,

$$\mathcal{F}_{\mathbb{C}}(e^{-t|\cdot|^2})(x) = \frac{\kappa_F}{t} e^{-\kappa_F^2 |x|^2/t}.$$

Fourier duality and then Tonelli's theorem on the nonnegative side give, in polar coordinates,

$$M(\sigma) = \kappa_F^{1-2\sigma} \frac{\Gamma(\sigma)}{\Gamma(1-\sigma)} \int_0^\infty W_0(r) r^{-\sigma} dr > 0.$$

The last integral is finite by annular support and positive because W_0 is nonnegative and nonzero. In particular $M(1/6) > 0$. The zeta residue is

$$\rho_S = \operatorname{Res}_{s=1} \zeta_F^S(s) = \frac{\pi}{3\sqrt{3}} \prod_{\mathfrak{p} \in S} (1 - q_{\mathfrak{p}}^{-1}) > 0.$$

Indeed, lattice counting and division by the six units give $\#\{\mathfrak{a} : N\mathfrak{a} \leq X\} = \pi X/(3\sqrt{3}) + O(\sqrt{X})$; partial summation gives the undeleted residue, and deletion supplies the displayed finite factors. Thus $c_S = \widehat{W}_1(1)M(1/6)\rho_S^2/6 > 0$. Both zeta residues contribute to this constant. At the balanced scales we write

$$I_\eta(Z) = I_\eta(Z^{1/2}, Z^{1/2}, Z), \quad J_\eta(Z) = c_S^{-1} I_\eta(Z). \quad (7)$$

Define $H_\eta(s)$ by substituting $(x, w, z) = (s, 1, 1/6)$ into the local correction factors and taking their product. Thus it is the restriction $\mathcal{H}_{\eta,1}(s, 1, 1/6)$ on the continuation domain established in Subsection 4.2. The calculation below proves convergence of this restricted product directly. For a prime $\mathfrak{p} \notin S$, put $Q = N\mathfrak{p}$ and $\alpha(\mathfrak{p}) = \alpha(p)$ for its primary generator p . With $t = Q^{-1}$, $D = \eta(\mathfrak{p})Q^{-s}$, and $R = \overline{\alpha(\mathfrak{p})}^6 \eta(\mathfrak{p})^6 Q^{3-6s}$, the local factor simplifies to

$$H_{\mathfrak{p}}(s) = \frac{1-t^2}{1-R} \left(1 + \frac{t(D-R)}{1-D} \right). \quad (8)$$

For example, the $j = 0$ line of the local table has $V = W = t$, $\rho = 1$, and $\mathcal{R} = R$. Its full factor is $P_p = (1+t)/(1-t) + (1+t)(R-D)/(1-R)$; multiplying by $(1-t)^2/(1-D)$ gives (8).

On $\operatorname{Re} s \geq 11/12$ we have $|D| \leq Q^{-11/12} < 1/2$ and $|R| \leq Q^{-5/2} < 1/2$, since every good prime has $Q \geq 7$. Subtracting one in (8) consequently gives the uniform bound

$$|H_{\mathfrak{p}}(s) - 1| \leq 4(Q^{-2} + Q^{-23/12} + Q^{-5/2}) =: a_Q.$$

All three prime sums converge. Choose the initial cutoff P_0 so that $\sum_{N\mathfrak{p} > P_0} a_{N\mathfrak{p}} \leq \log(3/2)$. This choice is independent of the target and of S . The product converges locally uniformly to a holomorphic function on $\operatorname{Re} s > 11/12$, and the finite-product inequality $|\prod(1+h_p) - 1| \leq \exp(\sum |h_p|) - 1$ passes to its limit, giving

$$|H_\eta(s) - 1| \leq \frac{1}{2} \quad (\operatorname{Re} s > 11/12). \quad (9)$$

Additional conductor-prime deletions only reduce the majorant sum.

The principal signal. With this correction define

$$f_\eta(Z) = \frac{1}{2\pi i} \int_{(2)} Z^{x-2/3} e^{(x-5/6)^2} \frac{H_\eta(x)}{L_F^S(x, \eta)} dx. \quad (10)$$

The integral is absolutely convergent for every $Z > 0$, since the Gaussian decays on the line $\operatorname{Re} x = 2$ and both H_η and the reciprocal Euler product are bounded there. The double residue of the principal row will be shown to equal $c_S f_\eta(Z)$ in Subsection 4.3.

3 The low estimate

Reflection replaces the cubic coefficients by moving quadratic characters χ_R^3 . We apply the quadratic large sieve with the remaining prime-power factors frozen, then sum those factors.

3.1 Reflection and the quadratic large sieve

Use the completed row (3) with $V = V_G$, so $\widehat{V}(t) = e^{t^2}$. The dual variables introduced by reflection may meet S .

For $K, N \geq 1$, the quadratic Hecke large sieve [3, Definition 1 and Theorem 1.1], after a fixed finite ray subdivision, gives

$$\sum_{q_R \leq K}^{\text{sf}} \left| \sum_{q_n \leq N}^{\text{sf}} a_n \chi_R(n)^3 \right|^2 \ll_\varepsilon (KN)^\varepsilon (K+N) \sum_n |a_n|^2. \quad (11)$$

Both indices in this inequality are squarefree and prime to S ; the coefficients a_n are arbitrary complex numbers. Here is the conductor check for the cited theorem. On ideals prime to λR , let $\theta_R((x)) = (x_{\text{prim}}/R)_2$, where x_{prim} is the primary generator of (x) . Write ϵ_λ for the nontrivial character of $(\mathcal{O}/\lambda)^\times$, and put $e(R) = (q_R - 1)/2 \pmod{2}$. Adjusting x to its primary generator gives

$$\theta_R((x)) = \epsilon_\lambda(x)^{e(R)} (x/R)_2.$$

Indeed, the powers of ω are squares, and the sign of the adjusting unit is determined by $x \pmod{\lambda}$. By CRT, the primitive inducing character has exact conductor $R\lambda^{e(R)}$ and trivial infinite type. When $e(R) = 0$ its factor at λ is absent. A fixed ray subdivision, including the residue modulo 4, fixes $e(R)$. Thus coprime members in one class have primitive product conductor exactly their product: the two factors at λ cancel. The reciprocity factor is the cube of the finite bicharacter R in (1), by the sextic reciprocity law proved in Appendix A. These are the Hecke-family hypotheses of the cited theorem; reciprocity, or transpose duality, gives the displayed orientation. On the indicated support the primitive characters have exactly the values $\chi_R(n)^3$, including zeros. Restricting the outer row set decreases the nonnegative left-hand side; a fixed coefficient-support restriction is imposed by setting the excluded coefficients to zero before applying the inequality.

Lemma 3 (Weak reflection). *Factor a nonzero row as $(m) = PsR$, where P is the product of the prime-power factors with exponent at least two, s is squarefree and supported on S , and R is squarefree and prime to SP . Freeze P, s and the unit of m . The completed row is a sum of $O_{S,\nu}(2^{\omega(P)})$ expressions of the form*

$$\zeta_R \sum_{\mu \neq 0} \frac{d(\mu) \alpha(\mu) \vartheta(\lambda^4 \mu)}{q_\mu^{1/2}} \chi_R(\lambda^4 \mu)^3 \prod_{p|D} B_p(\lambda^4 \mu) V^\sharp \left(\frac{q_\mu Z}{q_{c_F D R}^2} \right). \quad (12)$$

Here $\omega(P)$ counts the distinct prime divisors of P , $D \mid \text{rad}(P)$ is prime to S , c_F has finitely many possibilities depending only on the fixed data, and $|\zeta_R| \leq 1/81$. After a fixed finite ray subdivision of R , both d and ϑ are independent of R . The function $d(\mu)$ is the conjugate cusp coefficient defined in Appendix B; ϑ is its fixed additive phase. The frozen factors are independent of R and satisfy $|B_p(x)| \leq q_p^{1/2}$. The cusp support is

$$\mu = u\lambda^k n b^3, \quad k \geq -4, \quad n \text{ squarefree}, \quad |d(\mu)| \leq 27 \cdot 3^{k/6} q_b^{1/2}.$$

The dual n, b are primary, may share primes, and need not avoid S . For every $A \geq 0$ and integer $j \geq 0$, the kernel satisfies

$$|(x\partial_x)^j V^\sharp(x)| \ll_{A,j} \min(1, x^{-A}) \quad (x > 0).$$

These bounds are uniform in the rows and reflection branches.

The proof, including the fixed cusp choices and the local zero masks, is given in Appendix B.

3.2 Balanced row energy

Lemma 4 (Balanced energy). *For every fixed $C > 0$, every fixed finite family of ν , and $\varepsilon > 0$, uniformly for $Z \geq 2$,*

$$\sum_{0 < q_m \leq CZ} |C_V(Z; m, \nu)|^2 \ll Z^{1+\varepsilon}.$$

Proof. Freeze P, s , the unit of m , a reflection branch and a ray sector. Then each ideal R specifies at most one row. Put $p_0 = NP$ and $D_0 = ND \leq \sqrt{p_0}$; the inequality follows because every prime in P has exponent at least two. Place R in an annulus $q_R \asymp K \leq K_{\max}$, where $K_{\max} \ll Z/p_0$. Use smooth dyadic partitions, including the bounded ranges, for $q_n \asymp N$ and $q_b \asymp B$. For a fixed unit and $k \geq -4$, Lemma 3 bounds the coefficient by a fixed constant times $D_0^{1/2} 3^{-k/3} N^{-1/2} B^{-1}$. After this normalization, the remaining column coefficient is bounded by one and independent of R . The factor ζ_R is a bounded row scalar and can be removed from the positive row norm. Keep the kernel and the three annular cutoffs as one smooth profile in $(q_R/K, q_n/N, q_b/B)$. Simultaneous logarithmic Fourier inversion has a single common coefficient density with uniformly bounded L^1 norm. Indeed, in logarithmic coordinates its kernel is $V^\sharp(t_0 \exp(\xi_n + 3\xi_b - 2\xi_R))$, multiplied by three fixed compactly supported cutoffs. Its derivatives through order four are uniformly bounded for every $t_0 > 0$, by the Euler-derivative bounds. Applying $(1 - \Delta)^2$ bounds its Fourier transform by $C(1 + |\mathbf{t}|^2)^{-2}$, which is integrable on $\mathbb{R}^3$. The constant is independent of P and every block parameter. Apply Minkowski to this common density before enlarging any positive row norm.

In each separated mode, freeze b first. The factors $\chi_R(u\lambda^{k+4})^3$ and $\chi_R(b)^3$ are then bounded row factors, including their zeros. Apply (11) in the squarefree n variable, separating its finitely many squarefree S -parts: writing $n = n_S n_0$ makes $\chi_R(n_S)^3$ another bounded row factor and leaves squarefree n_0 prime to S . A condition involving the now fixed b is a fixed coefficient-support restriction. With coefficients bounded by one, the squared norm is $\ll_\delta (KN)^\delta (K + N)N$ for every $\delta > 0$. Minkowski over the $O(B)$ choices of b and the displayed normalization give

$$\|\text{dual block}\|_2^2 \ll_\delta (KN)^\delta D_0 3^{-2k/3} (K + N). \quad (13)$$

Here is a uniform truncation of the dual sum. For a fixed row let

$$L_R = \frac{q_{cF}^2 D_0^2 q_R^2}{Z} \ll Z.$$

Fix $\tau > 0$ and discard whole blocks with $3^k N B^3 > Z^\tau D_0^2 K^2 / Z$. On such a block the kernel argument q_μ / L_R is at least a fixed positive multiple of Z^τ . The cusp support lies in $\lambda^{-4} \mathcal{O} \setminus \{0\}$, and $|d(\mu)| / q_\mu^{1/2} \ll 1$. Lattice counting gives

$$\sum_{\mu \in \lambda^{-4} \mathcal{O} \setminus \{0\}} (1 + q_\mu / L_R)^{-2} \ll 1 + L_R.$$

For any fixed $A > 0$, rapid decay therefore bounds the discarded part of one branch in one row by

$$O_A(D_0^{1/2} (1 + L_R) Z^{-A\tau}) = O_A(Z^{5/4 - A\tau}).$$

The dyadic partitions have bounded overlap, and the representation $\mu = u \lambda^k n b^3$ on this support is unique. There are $O(Z)$ rows, and the number of branches per row is $O_\eta(Z^\eta)$ for every $\eta > 0$. Thus the total discarded row norm is $O_{A,\eta}(Z^{7/4 + \eta - A\tau})$, which is smaller than any prescribed negative power by choosing A sufficiently large. Discarding whole blocks preserves the common coefficient profile used above.

For a retained block,

$$N \ll \frac{Z^\tau D_0^2 K^2}{3^k Z B^3}, \quad 3^k N B^3 \ll Z^{1+\tau}, \quad K N \ll Z^{2+\tau}.$$

There are $O(\log(2Z))$ choices for each annulus and for k . On taking square roots in (13), the two terms have k factors $3^{-k/3}$ and $3^{-5k/6}$, respectively; both are summable for $k \geq -4$. The row-space triangle inequality over the remaining annuli consequently gives, for one fixed branch,

$$\sum_R |\text{branch}(R)|^2 \ll_{\delta,\tau} Z^{\tau + (2+\tau)\delta} (\log(2Z))^6 D_0 \left(K_{\max} + \frac{D_0^2 K_{\max}^2}{Z} \right).$$

Since $D_0 \leq \sqrt{p_0}$, the last factor is

$$\ll Z \left(\frac{D_0}{p_0} + \frac{D_0^3}{p_0^2} \right) \ll Z p_0^{-1/2}.$$

Choose τ, δ small enough that the preliminary losses, including the logarithm, are at most $Z^{\varepsilon/3}$. If there are $B(P)$ branches, rowwise Cauchy and summation over branches cost at most $B(P)^2 \ll_{\varepsilon,S,\nu} p_0^{\varepsilon/3}$. Finally every powerful ideal is uniquely $\mathfrak{x}^2 \mathfrak{y}^3$ with $\mathfrak{y}$ squarefree. Hence

$$\#\{P : N P \leq X\} \ll X^{1/2} \sum_{\mathfrak{y}} (N \mathfrak{y})^{-3/2} \ll X^{1/2}, \quad \sum_{N P \leq C Z} (N P)^{-1/2} \ll \log(2Z).$$

The convergent ideal sum uses the $O(T)$ ideal count from Section 1. Sum the fixed- P bounds and absorb this final logarithm into $Z^{\varepsilon/3}$. The S -parts, units and ray sectors have only finitely many possibilities. This proves the lemma. $\square$

3.3 From row energy to the probe

The additive norm. For $Y, Q \geq 1$, the Gauss expansion writes $A_{m,\sigma,v} = \sum_{s,d} a_{s,d} e(-md/s)$, where d is a unit modulo s , $q_s \asymp Y$, and $|a_{s,d}| \ll Y^{-3/2}$. These fractions are distinct: equality modulo $\mathcal{O}$ implies $s \mid s'$ and $s' \mid s$, hence equality of the primary denominators and residue classes. Distinct fractions modulo $\mathcal{O}$ are separated by $\delta \gg Y^{-1}$, since their difference has a nonzero integral

numerator and denominator of absolute value $O(Y)$. There are $O(Y^2)$ pairs, so $\sum |a_{s,d}|^2 \ll Y^{-1}$, uniformly in real v : its contribution $(q_s/Y)^{iv}$ has modulus one. To bound their row norm, majorize the ball $q_m \leq CQ$ by a fixed multiple of $e^{-cq_m/Q}$. Poisson summation gives a Gram kernel bounded in absolute value by $C'Q \sum_{n \in \mathcal{O}} e^{-c'Q|n-z_j+z_k|^2}$. The lifts to $\mathbb{C}$, including all lattice translates of each fraction, are still separated by a constant times Y^{-1} . They have at most $O(1+r^2/\delta^2)$ points in a radius- r disk; summing Gaussian shells bounds every matrix row sum by $O(Q+\delta^{-2})$. The elementary Schur bound therefore gives

$$\sum_{q_m \leq CQ} |A_{m,\sigma,v}(Y)|^2 \ll \frac{Q+Y^2}{Y}. \quad (14)$$

Apply the exact separation (4) with $X = Y = Z^{1/2}$, so $Q_0 = q_{b_*}Z$. The factor B is a fixed finite combination of completed rows (3). The fixed compact support in the low formula restricts $q_m \ll Z$. Thus (14) gives squared additive norm $O(Z^{1/2})$, uniformly in the Mellin height. Cauchy, Lemma 4, and the integrability of $\widehat{W}_0(iv)$ now yield $|I_\eta(Z^{1/2}, Z^{1/2}, Z)| \ll Z^{-1/2} Z^{1/4} Z^{1/2+\varepsilon} = Z^{1/4+\varepsilon}$. Division by the fixed positive principal-residue normalizer c_S of Subsection 2.5 gives the same bound for J_η .

4 The comparison estimate

Throughout this section assume $\beta_* > b = 47/48$, and fix the target η and the data of Section 2. Implied constants may depend on these fixed data and on the explicitly chosen contour buffers.

4.1 Buffered rows and zero density

We bin rows by their rightmost zero up to a finite height. Buffered contours control reciprocals within each bin, and zero density bounds the number of rows requiring the costlier contours.

Use the fixed families Θ and $\mathcal{X}_u$ from Subsection 2.4, retaining their prescribed zero extensions. Lemma 2 supplies the conductor and multiplicity bounds used below.

Lemma 5 (A zero-free disk). *Let ψ be a primitive nonprincipal finite-order Hecke character of conductor norm Q . Fix $0 < e < 10^{-3}$, $51/100 \leq a \leq 1$, and put $R_j = 2 - a - je$. If the disk $|s - (2 + it)| < R_2$ has no zero, then for every $\delta > 0$,*

$$|L(s, \psi)| + |L(s, \psi)^{-1}| \ll_{e,\delta} \{2Q(3 + |t|)^2\}^\delta \quad (|s - (2 + it)| \leq R_6).$$

The constants are uniform in a, Q, t, ψ .

Proof. For a primitive finite-order character the standard completion and functional equation are

$$\begin{aligned} \Lambda(s, \psi) &= (3Q)^{s/2} (2\pi)^{-s} \Gamma(s) L(s, \psi), \\ \Lambda(s, \psi) &= \epsilon_\psi \Lambda(1 - s, \bar{\psi}), \quad |\epsilon_\psi| = 1. \end{aligned}$$

This equation, Stirling's formula, and Euler convergence give the deliberately weak strip estimate $L(\sigma + iv, \psi) \ll Q^{3/5} (3 + |v|)^2$ for $-1/10 \leq \sigma \leq 11/10$; farther right use Euler convergence. Phragmén–Lindelöf follows from the usual theta integral, or by multiplying the normalized strip function by $e^{\delta(s-1/2)^2}$ and letting the height tend to infinity before $\delta \downarrow 0$.

Choose $g = \log L$ in the zero-free disk, normalized by the Euler logarithm at its center. Write $\mathcal{C} = 2Q(3 + |t|)^2$. The disk of radius R_3 is compactly contained in the assumed zero-free disk. The growth estimate bounds $\operatorname{Re} g = \log |L|$ there by $O(\log \mathcal{C})$, and the Euler logarithm gives

$|g(2 + it)| \ll 1$. Borel–Carathéodory on radii R_3, R_4 therefore gives $\max_{R_4} |g| \ll_e \log \mathcal{C}$. On the inner disk of radius $r_0 = 49/100$, Euler convergence gives $|g| \ll 1$. Three circles therefore gives

$$\max_{R_6} |g| \ll_e (\log \mathcal{C})^{\theta_e}, \quad \frac{\log(R_6/r_0)}{\log(R_4/r_0)} \leq \theta_e < 1.$$

Indeed $r_0 < R_6 < R_4 < R_3 < R_2$; the strict uniform inequality follows by compactness in $a \in [51/100, 1]$ for fixed e . Exponentiating g and $-g$, and using $\exp(C_e(\log \mathcal{C})^{\theta_e}) \ll_{e,\delta} \mathcal{C}^\delta$, proves the result. $\square$

The products of deleted factors, and their inverses, are $O_{\sigma_0,\delta,S}((N \operatorname{rad}(u))^\delta)$ on any fixed half-plane $\operatorname{Re} s \geq \sigma_0 > 0$. Indeed each logarithm is $O_{\sigma_0}((N \mathfrak{p})^{-\sigma_0})$, which is at most $\delta \log N \mathfrak{p}$ outside a fixed finite set. On $\operatorname{Re} s \geq -v$ for fixed $v \geq 0$, only the direct product is needed, and it costs $(N \operatorname{rad}(u))^{v+\delta}$.

Global reciprocal bounds. For a principal denominator apply the same disk proof to the holomorphic regularization $\mathcal{Z}(s) = (s-1)\zeta_F(s)/(s+1)$ on $\operatorname{Re} s > 0$, with its removable value at one, and then use $1/\zeta_F = (s-1)/(s+1) \cdot [(s-1)\zeta_F/(s+1)]^{-1}$. The regularized value at one is nonzero. Its logarithm is uniformly bounded on the inner Euler disk, and it has polynomial growth on the larger disk. Given a fixed $\kappa > 0$, apply the disk argument with $a = \beta_*$ and a buffer smaller than $\kappa/6$; these disks are zero-free at every height. Euler convergence covers the part farther to the right. After restoring the deleted factors, for every $\delta > 0$ this gives

$$|L_{\operatorname{orig}}(\sigma + it)^{-1}| \ll_{\kappa,\delta,S,\Theta} q_u^\delta (3 + |t|)^\delta \quad (\sigma \geq \beta_* + \kappa). \quad (15)$$

At a principal pole the reciprocal is interpreted by its holomorphic continuation, with value zero. The same estimate holds for unit rows, whose conductors and deleted primes are fixed.

Lemma 6 (Fixed-height bins). *Fix $0 < e < 10^{-3}$ and assume $\beta_* \geq 51/100$. For $U \geq 1$ and $T \geq 3$, each nonunit row with $U \leq q_u < 2U$ can be assigned one of $O_e(1)$ labels $a \in [51/100, \beta_*]$ with the following properties. The bounds hold for every $\delta > 0$, uniformly in the row, label and height; their constants may depend on e, δ and the fixed data.*

1. *If $a > 51/100$, some inducing function in $\mathcal{X}_u$ has a zero with real part at least a and imaginary part of absolute value at most $2T$.*
2. *For every presentation in $\mathcal{X}_u$,*

$$|L_{\operatorname{orig}}(\sigma + it)| + |L_{\operatorname{orig}}(\sigma + it)^{-1}| \ll U^\delta (3 + T)^\delta \quad (\sigma \geq a + 6e, |t| \leq T).$$

3. *On the reflected line,*

$$|L_{\operatorname{orig}}(1 - a - 6e + it)| \ll U^{a-1/2+12e+\delta} (3 + T)^A \quad (|t| \leq T),$$

where A is fixed.

Proof. All inducing characters in a nonunit row are nonprincipal by Lemma 2. Let M_u be the maximum of $51/100$ and the real parts of all zeros of the finite inducing family with $\operatorname{Re} \rho \geq 51/100$ and $|\Im \rho| \leq 2T$. Put $a = 51/100 + e[(M_u - 51/100)/e]$. Euler convergence bounds these real parts by one, so the maximum exists by discreteness of zeros in a compact rectangle. Thus $a \leq M_u < a + e$ and $a \leq \beta_*$. If the label is above the floor, the maximum is supplied by an actual zero, proving the first claim.

For $|t| \leq T$, the disk centered at $2 + it$ of radius $2 - a - 2e$ has real part greater than $a + 2e > M_u$ and height less than $T + 3/2 < 2T$. Apply Lemma 5, the conductor bound, and the deleted-factor estimate. The disk conclusion covers the real segment from $a + 6e$ to $2 + R_6$, and Euler convergence covers larger real parts. Shrinking preliminary losses gives the stated direct and reciprocal bounds. The functional equation reflects the other line to $a + 6e - it$ for the conjugate primitive character, which also belongs to the inducing family because Θ is closed under conjugation. It costs $Q^{a-1/2+6e}$ and a fixed height power. Its deleted factors cost at most $U^{6e+\delta}$. This proves the last bound. $\square$

The published general Hecke zero-density theorem [4, Lemma 3.3], restricted to angular type zero over our degree-two field, states, uniformly for $1/2 \leq \sigma < 1$, $Q > 1$, and $T > 2$,

$$\sum_{N \nmid \mathfrak{f} \leq Q} \sum_{\psi \bmod \mathfrak{f}}^* N(\sigma, T, \psi) \ll (Q^2 T^2)^{h(\sigma)(1-\sigma)} (\log QT)^B, \quad h(\sigma) = \min \left\{ \frac{3}{2-\sigma}, \frac{2}{\sigma} \right\} \leq \frac{5}{2}.$$

Here N counts zeros with real part at least σ and $|\Im \rho| \leq T$, with multiplicity, and the star means primitive. The published theorem also sums over angular parameters; retaining only angular type zero decreases its nonnegative left-hand side. For $a > 51/100$, assign to each row a character and zero supplied by Lemma 6. Its conductor is $O_{S,\Theta}(U)$, and a fixed primitive character occurs in at most twelve physical rows by Lemma 2. Apply the density theorem at height $2T$, use $2h(a) \leq 5$, and absorb its logarithm into $U^\delta(3+T)^A$. Let $\mathcal{B}_a$ denote the set of nonunit rows with $U \leq q_u < 2U$ assigned label a . Together with the trivial $O(U)$ lattice count, this gives

$$\#\mathcal{B}_a \ll U^{R(a)+\delta}(3+T)^A, \quad R(a) = \min\{1, 5(1-a)\}. \quad (16)$$

The floor uses only the trivial count. At a possible label $a = 1$, take $\sigma \uparrow 1$ in the uniform theorem for fixed conductor and height ranges.

4.2 Euler-product domains and contour tails

Local domains. Write $x_r = \operatorname{Re} x$, $w_r = \operatorname{Re} w$, and $z_r = \operatorname{Re} z$. Here is the local verification needed for all contour paths. For a good prime of norm Q , put

$$d = \eta(\mathfrak{p})Q^{-x}, \quad h = Q^{-w}, \quad E = dh, \quad V = Q^{-6z}, \quad R = a_{\mathfrak{p}}^2 Q^{4-6x-6z},$$

where $a_{\mathfrak{p}}$ is the modulus-one arithmetic phase defined in Subsection A.3. At a prime $\mathfrak{p} \nmid u$, $\rho = \chi_{\mathfrak{p}}(u)$ satisfies $\rho^6 = 1$, and the finite Euler identity is

$$H_{\mathfrak{p}}(\rho) = \frac{1}{1-R} \left(1 + \frac{B_0 + dV\rho^{-1} + A_1\rho + A_2\rho^2}{1-d\rho^{-1}} \right), \quad (17)$$

$$B_0 = E(1-QV) - R/Q, \quad A_1 = h[-V + R/Q + E(Q-1)V], \quad A_2 = -R(1-V)h^2.$$

For $j = v_{\mathfrak{p}}(u) \geq 1$, reset $\rho = \chi_{\mathfrak{p}}(u/\pi_{\mathfrak{p}}^j)$ for the fixed primary generator $\pi_{\mathfrak{p}}$, so that $\rho^6 = 1$; the full ramified factors are

$$H_{\mathfrak{p},1} = \frac{1 - R/Q + E(1-QV)}{1-R},$$

$$H_{\mathfrak{p},j} = \frac{1 - R/Q - E(Q-1) + (1-V)J_j}{1-R} \quad (j \geq 2), \quad (18)$$

These formulas follow by multiplying the full factor $P_{\mathfrak{p}}$ of Subsection A.3 by $(1-V)(1-\rho h)/(1-d\rho^{-1})$ at unramified primes, and by $1-V$ at ramified primes. Here $J_1 = E$ and $J_2, \dots, J_5$ are given in

(A.4), with $|a_p| = |b_p| = 1$; only $1 \leq j \leq 5$ occurs. The factor $V^{1_{j \leq 1}}$ accounts for the term $-E(Q-1)$ for every $j \geq 2$.

On the domain

$$x_r \geq \frac{51}{100}, \quad z_r \geq \frac{17}{50}, \quad w_r \geq -\frac{1}{100}, \quad x_r + w_r \geq 1 + \delta, \quad \delta > 0, \quad (19)$$

we have $|d| < 1$ and $|R| \leq Q^{-11/10}$. In (17), the unit monomials have decay exponent greater than one: E has exponent at least $1 + \delta$, and the other monomials have a fixed margin. Their decay exponents have the following lower bounds:

monomial	lower bound	monomial	lower bound
R	$11/10$	E	$1 + \delta$
EQV	$51/25 + \delta$	R/Q	$21/10$
dV	$51/20$	hV	$203/100$
hR/Q	$259/100 + \delta$	$hEQV$	$203/100 + \delta$
Rh^2	$52/25 + 2\delta$		

The remaining terms contain an extra factor V or Q^{-1} . In (18), the additional $E(Q-1)$ has exponent at least δ . The monomials in J_2, J_3, J_4, J_5 have respective lower bounds $3/100, 1/50 + \delta$ and $1/25, 3/100 + \delta$, and $3/50$. Thus the unramified factors are $1 + O_\delta(Q^{-1-\min(\delta, 1/10)})$, while the ramified factors are $1 + O_\delta(Q^{-\min(\delta, 1/50)})$. The denominators $1 - d\rho^{-1}$ and $1 - R$ do not vanish. Normal convergence at the unramified primes and the elementary divisor bound at the finitely many ramified primes prove holomorphy on a neighborhood of each fixed real box in this domain and the bound

$$\mathcal{H}_{\eta,u} \ll_{\delta,\epsilon} q_u^\epsilon \quad \text{on every fixed real box in (19),} \quad (20)$$

uniformly in all heights.

We also use the separate principal box

$$x_r \geq \frac{15}{16}, \quad w_r \geq \frac{3}{4}, \quad z_r \geq \frac{1}{8}. \quad (21)$$

Here $|d| \leq Q^{-15/16}$ and $|R| \leq Q^{-19/8}$. All unit monomials in (17) have exponent at least $23/16$; the limiting one is EQV . Consequently $\mathcal{H}_{\eta,1} \ll 1$ on this whole box. The unramified decay exponents increase with x_r, w_r, z_r , so the bound holds throughout the principal contour paths. For the separate large-row domain $x_r, w_r, z_r \geq 2$, the ramified term $E(Q-1)$ has decay exponent at least three, and every J_j has exponent greater than one. Thus $\mathcal{H}_{\eta,u} \ll 1$ uniformly for all rows on that domain.

A common tail bound. On fixed real strips, $\widehat{W}_1$ is Schwartz, and M is holomorphic and Schwartz on compact positive real strips. For M , integrate by parts in the logarithmic radial variable; the endpoint at zero is integrable because $z_r > 0$. The independent height coordinates are $(t_x + t_z, t_z, t_w)$. Hence for every L ,

$$|e^{(x+z-1)^2} M(z) \widehat{W}_1(w)| \ll_L (1 + |t_x| + |t_z| + |t_w|)^{-L}.$$

If the arithmetic factor on a specified domain is bounded by $Z^B(1 + |\mathbf{t}|)^A$, integration outside the box $|t_x|, |t_z|, |t_w| \leq T/4$, or on a horizontal face at height $T/4$, is $O_N(Z^B T^{-N})$: take $L > N + A + 3$. The same proof applies to the two variables remaining after the w residue. The finite real length of a horizontal join costs only a constant. Increasing N changes only external test seminorms, not the arithmetic growth orders A or B . Before a reciprocal contour is moved into a finite-height region, (15) supplies its all-height bound. Afterwards, only faces and segments inside that region are used.

4.3 Contour shifts and row estimates

Use the balanced specialization (7) in (5)–(6), with principal signal (10). The physical probe is independent of the contour height chosen below.

Nonunit rows below the large-row cutoff. For $Z \geq 2$, take a dyadic $U \geq 1$ and write $U = Z^d$ with $0 \leq d \leq 501/1000$. Fix $T \geq 3$ and a bin $\mathcal{B} = \mathcal{B}_a(U, T)$ of nonunit rows with $U \leq q_u < 2U$ before moving any variable. The sets in this partition may depend on T ; the original sum over all rows does not. Set $z_0 = 17/50$ and choose a fixed small $0 < e < 10^{-3}$. Starting in (5)–(6), move z to z_0 and x to $\beta_* + 20e$, with $w = 3$; then move w to $1 - a - 6e$. Throughout, $x_r + w_r \geq 1 + 14e$, the numerator is nonprincipal, the reciprocal remains to the right of β_* , and $6z_r > 1$. The paths lie in (19), so no pole is crossed and the tail bound justifies the infinite horizontal limits.

While x is still global, discard the complement of $|t_x|, |t_w|, |t_z| \leq T/4$. Its cost is $O_N(Z^B T^{-N})$. Move only the retained x segment to $a + 16e$. Its rectangle has $a + 16e \leq x_r \leq \beta_* + 20e$ and $|t_x| \leq T/4$, so it lies strictly inside the region covered by Lemma 6. On the new segment $x_r + w_r = 1 + 10e$, and the same lower bound holds on the two horizontal joins. To estimate those joins, extend the absolute integrals in (t_w, t_z) to $\mathbb{R}^2$ and use global polynomial numerator bounds. This is valid because the reciprocal depends only on the still-buffered x , not on w or z . The face bound again gives $O_N(Z^B T^{-N})$. The global polynomial conductor costs of the numerators are absorbed into Z^B , since $U \leq Z^{501/1000}$. These growth orders are uniform in the labels and dyads. No moved reciprocal axis is extended beyond its zero-free rectangle. Write $E_{d,a}(Z, T)$ for the retained integral on the new vertical segment. The original bin contribution equals this integral plus the discarded tails and horizontal joins just bounded.

On the retained contour the buffered functional equation gives $L_F^S(w, \psi_u) \ll U^{a-1/2+12e+\epsilon}(1+T)^A$. The balanced outside power and the row weight are respectively

$$Z^{x_r+z_r/2+w_r/2-5/4} = Z^{a/2-3/4+17/100+13e}, \quad q_u^{-z_r} \asymp U^{-17/50}.$$

The reciprocal and (20) cost arbitrarily small powers of U and T , and the external kernel has a bounded integral over the three heights. Thus the conductor power is $U^{a-21/25+12e+\epsilon_1}$ for an arbitrarily small $\epsilon_1 > 0$. Since $U \leq Z^{501/1000}$, taking e and ϵ_1 sufficiently small for a prescribed $\epsilon > 0$ proves

$$|E_{d,a}(Z, T)| \ll_{\eta, \epsilon} Z^{a/2-3/4+17/100+\epsilon} U^{a-21/25} \# \mathcal{B}(1+T)^A. \quad (22)$$

Principal, other unit, and large rows. For the principal row first move to $(x_r, w_r, z_r) = (\beta_* + e, 1 + e, 1/6 + e)$, then move w to $3/4$, crossing its pole at 1. In that residue move z to $1/8$, crossing its pole at $1/6$; the unresidued integral keeps $z_r = 1/6 + e$. All paths lie in (21); M has no pole and the reciprocal stays global. The double residue is $c_S f_\eta(Z)$, moving the remaining x line to 2 without crossing a pole (also for principal η). The common tail bound applies before and after taking residues. For the other five units, the numerator is nonprincipal by Subsection 2.5; move through (19) to the contours below, crossing no pole. For large dyads $U > Z^{501/1000}$, move to $(2, 2, v)$ with fixed $v > 2$, within the absolute Euler domain.

The outside power $Z^{x_r+z_r/2+w_r/2-5/4}$, polynomial height bounds, and (20) give the following estimates, with a common factor Z^ϵ suppressed. The value $w_r = 1$ denotes a residue.

remainder	(x_r, w_r, z_r)	bound
E_w	$(\beta_* + e, 3/4, 1/6 + e)$	$Z^{C(\beta_*)-1/8+3e/2}$
E_z	$(\beta_* + e, 1, 1/8)$	$Z^{C(\beta_*)-1/48+e}$
$\sum_{u \in \mu_6 \setminus \{1\}} E_u $	$(\beta_* + e, 1/2, 17/50)$	$Z^{C(\beta_*)-49/300+e}$
$ E_U $	$(2, 2, v)$	$Z^{7/4+v/2} U^{1-v+\epsilon}$

(23)

In the last row the uniform all-row Euler bound from Subsection 4.2 and dyadic counting are used. Summing gives $\sum_{U > Z^{501/1000}} |E_U| \ll Z^{7/4+501/1000-v/1000+\epsilon'}$. With $v = 4000$ the exponent before ϵ' is $-1749/1000$, so $\epsilon' < 749/1000$ gives $O(Z^{-1})$. Initial absolute convergence and this summable bound justify summing the dyadic contour identities.

Consequently $J_\eta - f_\eta$ is the sum of the normalized retained bin integrals, the principal, other-unit and large-row remainders, and the finite-contour errors. Equation (23) and the tail estimates, after division by c_S , bound all but the retained bin integrals by

$$O_{\eta,\epsilon,N}(Z^{C(\beta_*)-m_0+\epsilon}(1+T)^A + Z^B T^{-N}).$$

The $O_e(\log(2Z))$ tail terms may be absorbed by increasing B once. The exponents A, B are independent of N ; we choose the losses, T , and N in the next proof.

4.4 Combining the estimates

The cutoff exponent $501/1000 > 1/2$ permits the large-row shift above. The proof below absorbs its extra cost and the contour, height, and logarithmic losses into $m_0 = 1/1200$, retaining $\delta_0 = m_0/4$.

Lemma 7. *Suppose $\beta_* > b$. For $Z \geq 2$, $|J_\eta(Z) - f_\eta(Z)| \ll_\eta Z^{C(\beta_*)-1/4800}$. The saving $1/4800$ is independent of the target.*

Proof. Insert (16) in (22). Before the adjustable losses and height factor, the exponent relative to $Z^{C(\beta_*)}$ is

$$E(d, a) = \frac{a}{2} - \beta_* + \frac{13}{150} + d \left(R(a) + a - \frac{21}{25} \right).$$

At $d = 1/2$, this is $a - \beta_* + R(a)/2 - 1/3$. The two cases give

$$a + \frac{R(a)}{2} - \frac{1}{3} \leq \begin{cases} 4/5 + 1/2 - 1/3 = 29/30, & a \leq 4/5, \\ 13/6 - 3a/2 \leq 29/30, & a \geq 4/5. \end{cases}$$

Since $\beta_* > 47/48$, we obtain $E(1/2, a) < -1/80$. Also $0 < R(a) + a - 21/25 \leq 1$ throughout $51/100 \leq a \leq 1$. Thus $E(d, a)$ increases with d , and throughout $0 \leq d \leq 501/1000$,

$$E(d, a) \leq E(1/2, a) + \frac{1}{1000} < -\frac{1}{80} + \frac{1}{1000} = -\frac{23}{2000} < -\frac{1}{100}.$$

This saving for all nonunit dyads below the cutoff is larger than m_0 . First choose the buffer e and all preliminary conductor losses, and absorb the logarithmic number of terms, so that the total real loss is at most $m_0/4$. This is possible because the combined buffer loss in a nonunit dyad is at most $(13 + 12 \cdot 501/1000)e$, which tends to zero with e . The principal and other-unit losses are smaller fixed multiples of e . These choices fix the arithmetic growth orders A, B before any tail order is chosen. The row count and contour comparison now give, for every fixed N ,

$$|J_\eta - f_\eta| \ll_{\eta,N} Z^{C(\beta_*)-3m_0/4}(1+T)^A + Z^B T^{-N}.$$

After fixing the target, choose $0 < \tau \leq m_0/[4(A+1)]$ and $T = 3Z^\tau$, so the hypothesis $T \geq 3$ in Lemma 6 always holds. The first term is $O_\eta(Z^{C(\beta_*)-m_0/2})$. Choose N so large that $B - N\tau < C(b) - m_0/4$. The second term is then $O_\eta(Z^{C(\beta_*)-m_0/4})$. The external-tail argument permits this late choice without changing the previously fixed orders. Since $m_0/4 = 1/4800$, the result follows. $\square$

5 Proof of the main theorem

5.1 Mellin continuation and the Hecke conclusion

Suppose $\beta_* > b$ and fix a primitive finite-order target η , including the principal character. Section 3, with $\varepsilon = 1/32$, gives $J_\eta(Z) \ll_\eta Z^{9/32}$. Since $C(b) - \delta_0 - 9/32 = 149/4800 > 0$, Lemma 7 gives

$$f_\eta(Z) \ll_\eta Z^{\beta_* - 2/3 - \delta_0} \quad (Z \geq 1). \quad (24)$$

Here continuity of the absolutely and locally uniformly convergent integral (10) extends the estimate from $Z \geq 2$ to $Z \geq 1$. For $0 < Z \leq 1$, shift that integral from $\operatorname{Re} x = 2$ to any fixed $M > 2$. On this strip H_η and $1/L_F^S(x, \eta)$ are holomorphic and bounded, by (9) and absolute Euler convergence. Gaussian decay removes the horizontal joins, giving $f_\eta(Z) \ll_{\eta, M} Z^{M-2/3}$.

Put $r_* = \beta_* - \delta_0$. The Mellin transform

$$F_\eta(s) = \int_0^\infty f_\eta(Z) Z^{-s+2/3} \frac{dZ}{Z}$$

is holomorphic on $\operatorname{Re} s > r_*$. Indeed, on a compact subset with real parts in $[\sigma_-, \sigma_+]$, its endpoint majorants are $Z^{r_* - \sigma_-} dZ/Z$ at infinity and $Z^{M - \sigma_+} dZ/Z$ at zero, where $M > \max(2, \sigma_+)$. These are integrable even after insertion of any fixed power of $|\log Z|$, justifying differentiation. To identify the transform, set $A_\eta(s) = e^{(s-5/6)^2} H_\eta(s)/L_F^S(s, \eta)$ for $\operatorname{Re} s > 1$. The defining signal integral gives

$$g_\eta(u) := e^{-4u/3} f_\eta(e^u) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{itu} A_\eta(2+it) dt.$$

The density is continuous and integrable by Gaussian decay. Also $g_\eta(u) = O_\eta(e^{(\beta_* - 2 - \delta_0)u})$ at $+\infty$ and $O_{\eta, M}(e^{(M-2)u})$ at $-\infty$, so $g_\eta \in L^1(\mathbb{R})$. Fourier inversion therefore gives $F_\eta(2+it) = A_\eta(2+it)$ for every real t . Since $r_* < 1$, both functions are holomorphic on $\operatorname{Re} s > 1$; the identity theorem identifies them there.

Now $r_* > b - \delta_0 = 4699/4800 > 11/12$, where $|H_\eta| \geq 1/2$. Thus $G_\eta(s) = e^{-(s-5/6)^2} F_\eta(s)/H_\eta(s)$ is holomorphic on $\operatorname{Re} s > r_*$ and satisfies $G_\eta L_F^S(\cdot, \eta) = 1$ there by meromorphic continuation. This excludes zeros of $L_F^S(s, \eta)$ in that half-plane; the principal pole at 1 does not affect the argument. The same r_* works for every target, although the constants may depend on it. As $1/2 < r_* < \beta_*$, the definition of the supremum supplies a primitive target and a zero with real part greater than r_* , without requiring attainment. Deleted Euler factors are nonzero on $\operatorname{Re} s > 0$ and hence retain this zero, a contradiction. Therefore $\beta_* \leq 47/48$.

Absolute Euler convergence covers $\operatorname{Re} s > 1$. For a character η modulo $\mathfrak{f}$ induced by the primitive η^* of conductor $\mathfrak{f}^*$,

$$L_F(s, \eta) = L_F(s, \eta^*) \prod_{\substack{\mathfrak{p}|\mathfrak{f} \\ \mathfrak{p} \nmid \mathfrak{f}^*}} (1 - \eta^*(\mathfrak{p})(N\mathfrak{p})^{-s}).$$

Since these finite factors are nonzero on $\operatorname{Re} s > 0$, they introduce no zeros there and preserve the principal pole. This proves the Hecke assertion of Theorem 1.

5.2 Transfer to Dirichlet functions

Let χ be primitive of conductor q , and let χ_{-3} be the quadratic character modulo 3. On ideals prime to $q\mathcal{O}$, the norm pullback $\eta(\mathfrak{a}) = \chi(N\mathfrak{a})$ is a finite-order Hecke character, possibly imprimitive: if $\alpha \equiv 1 \pmod{q\mathcal{O}}$, then $\alpha\bar{\alpha} \equiv 1 \pmod{q}$, so η is trivial on ray principal ideals. For fractional α ,

these congruences are interpreted locally. Define $\Sigma_{\mathbb{Q}} = \{p : p \mid 3q\}$ and let Σ_F contain all primes above them; these deletion sets are independent of S . With $\chi\chi_{-3}$ presented modulo $\text{lcm}(q, 3)$,

$$L_F^{\Sigma_F}(s, \eta) = L^{\Sigma_{\mathbb{Q}}}(s, \chi)L^{\Sigma_{\mathbb{Q}}}(s, \chi\chi_{-3}). \quad (25)$$

At $p \nmid 3q$ the split case has local factor $(1 - \chi(p)p^{-s})^{-2}$; in the inert case the identity is $(1 - \chi(p)^2p^{-2s}) = (1 - \chi(p)p^{-s})(1 + \chi(p)p^{-s})$. This proves the formula on $\text{Re } s > 1$, hence everywhere by meromorphic continuation. Its left side is zero-free on $\text{Re } s > 47/48$ by the Hecke conclusion and nonvanishing of the deleted factors.

Away from $s = 1$, both Dirichlet factors are holomorphic, so neither can vanish in this region. At 1 the same argument applies unless one character is principal. Primitivity and uniqueness of the primitive inducer show that the exceptions are exactly $\chi = 1$ and $\chi = \chi_{-3}$. The other factor is then induced by χ_{-3} , and

$$L(1, \chi_{-3}) = \sum_{n \geq 0} \left(\frac{1}{3n+1} - \frac{1}{3n+2} \right) = \int_0^1 \frac{dt}{1+t+t^2} = \frac{\pi}{3\sqrt{3}} > 0.$$

Partial summation with bounded character sums identifies the convergent series at 1 with $L(1, \chi_{-3})$; each paired summand is $\int_0^1 t^{3n}(1-t) dt$, so nonnegativity justifies the interchange. Nonzero finite Euler factors preserve this value's nonvanishing and the principal pole, excluding pole-zero cancellation. Finally, imprimitive Dirichlet functions differ from their primitive inducers by such finite factors. The conductor-one case gives the Riemann zeta function, completing Theorem 1.

A Arithmetic identities and Euler factors

This appendix proves the exact high representation (5)–(6) for the physical probe (2), with the definitions of Section 2.

A.1 Gauss sums and the fixed phase correction

For every odd nonzero c , including nonsquarefree c , the function Γ_q in (1) is the normalized quadratic Gauss sum: $\Gamma_q(c) = |c|^{-1} \sum_{x \bmod c} e(x^2/c)$. To verify this with the self-dual measure of Subsection 2.1, apply Poisson summation to $e(z^2/c)e^{-\pi\delta|z|^2}$: its transform is

$$\frac{|c|}{2\sqrt{r}} e^{-\pi\delta q_c |y|^2/(4r)} e(-cy^2/(4r)), \quad r = 1 + 3q_c\delta^2/16.$$

The phase on the original side is periodic modulo $c\mathcal{O}$; after replacing r by one, the transformed phase is periodic modulo $2\mathcal{O}$. Divide by $2/(\sqrt{3}\delta)$ and let $\delta \downarrow 0$. The original periodic mean tends to $q_c^{-1} \sum_{x \bmod c} e(x^2/c)$, and the transformed mean tends to $(2|c|)^{-1} \sum_{y \bmod 2\mathcal{O}} e(-cy^2/4)$. Hence

$$\Gamma_q(c) = \frac{1}{2} \sum_{y \bmod 2\mathcal{O}} e(-cy^2/4),$$

which is (1), using the representatives $0, 1, \omega, 1 + \omega$. The phase error from replacing r by one is $O_c(\delta^2 \sum_y |y|^2 e^{-c_1\delta|y|^2}) = O_c(1)$, where $c_1 > 0$ depends on c ; the amplitude and damping errors are $O_c(\delta)$. All vanish after the preceding division.

In the quotient of $(\mathcal{O}/4\mathcal{O})^\times$ by its subgroup of squares, the four classes $1, -1, \lambda, -\lambda$ have Γ_q -values $1, 1, i, -i$. Thus R is a symmetric sign-valued bicharacter; explicitly its value on $(-1)^e \lambda^f, (-1)^g \lambda^h$

is $(-1)^{eh+fg+fh}$. For primary c prime to S , cubic reciprocity gives $\chi_c(4) = (c/(-2))_3$, a character modulo 2. Consequently, for primary v, w prime to S and $p \notin S$,

$$G(vw) = G(v)G(w)R(v, w), \quad G(p^3) = \gamma_3(p), \quad R(p, p) = \gamma_3(p)^2 = \chi_p(-1). \quad (\text{A.1})$$

For the middle identity, $\chi_{p^3}(4) = 1$ and $\Gamma_q(p^3) = \Gamma_q(p)$, since p^2 is a square unit modulo $4\mathcal{O}$. At a prime the quadratic exponential sum equals the quadratic character Gauss sum, so $\Gamma_q(p) = \gamma_3(p)$. Also $\Gamma_q(p^2) = 1$, and the usual quadratic Gauss identity $\gamma_3(p)^2 = \chi_p(-1) \in \{1, -1\}$ gives the last equality. These formulas descend to the ray group modulo $36\mathcal{O}$: $\chi_c(4)$ depends only on $c \bmod 2$, and $\Gamma_q(c)$ only on $c \bmod 4$. A unit relating primary representatives of the same ray class is 1 mod 3, hence is 1. For later use we record the corresponding sextic reciprocity law: for coprime primary v, w prime to S ,

$$\chi_v(w) = R(v, w)\chi_w(v). \quad (\text{A.2})$$

First take distinct primary primes v, w . Cubic reciprocity makes the square of $\chi_v(w)/\chi_w(v)$ equal to one. The quadratic Gauss sum and CRT give $R(v, w) = \chi_v(w)^3\chi_w(v)^3$, which is the cube of that quotient. A sign equals its cube, proving the prime case. Multiplicativity of the residue symbols and the bicharacter property of R extend the formula to arbitrary coprime v, w , including nonsquarefree arguments.

The prime Gauss–Jacobi identities, with these phases, are

$$\gamma_2(p)^3 = -\alpha(p), \quad \gamma_1(p)\gamma_2(p) = -\alpha(p)G(p).$$

Indeed, write $Q = Np$ and $J(\psi, \psi) = \sum_{x \in \mathcal{O}/p} \psi(x)\psi(1-x)$, with $\psi = \chi_p^2$. Reduction modulo p makes the summand $x^{(Q-1)/3}(1-x)^{(Q-1)/3}$; its degree is less than $Q-1$, so the finite-field sum is zero. Thus $p \mid J$. The Gauss–Jacobi identity gives $|J| = \sqrt{Q}$. On $\mathbb{F}_Q \setminus \{0, 1\}$, the order-three map $x \mapsto 1/(1-x)$ preserves the summand. Its two fixed points satisfy $x(1-x) = 1$, and all other orbits have size three. Hence $J \equiv 2 \equiv -1 \pmod{3}$. Since $J/p \in \mathcal{O}$ has norm one, it is a unit. The congruence $J \equiv -1 \pmod{3\mathcal{O}}$ and $p \equiv 1 \pmod{3\mathcal{O}}$ force that unit to be -1 , so $J = -p$. Since $g_{\chi_p^2}(p, 1)^2 = Jg_{\chi_p^4}(p, 1)$ and $g_{\chi_p^2}(p, 1)g_{\chi_p^4}(p, 1) = Q$, division by $Q^{3/2}$ gives $\gamma_2(p)^3 = -\alpha(p)$. For $\chi = \chi_p$, the substitutions $x = (1+y)/2$ and $y \mapsto y/(y-1)$ in the Jacobi sums give

$$J(\chi^2, \chi^2) = \overline{\chi(4)}^2 J(\chi^3, \chi^2) = \overline{\chi(4)}^2 J(\chi, \chi^2).$$

Since $\chi(4)^3 = 1$, this and $g_\chi g_{\chi^2} = J(\chi, \chi^2)g_{\chi^3}$ give the second formula. Writing $\mu(c)$ for the ideal Möbius function, CRT then gives

$$\gamma_2(c) = \frac{\mu(c)\alpha(c)G(c)}{\gamma_1(c)}, \quad \gamma_1(c) = \prod_{p|c} \gamma_1(p) \prod_{p < r, p, r|c} \chi_p(r)\chi_r(p). \quad (\text{A.3})$$

Here (A.3) is used only for squarefree c . For an arbitrary primary index A prime to S , $G(A)$ is still defined by the finite calibration (1), not by a sextic Gauss sum at modulus A .

A.2 Poisson summation and the moving-prime phases

All ideal variables in this subsection are represented by their primary generators and avoid S , and c is squarefree. For a nonzero sixth-power-free u prime to S , including its unit, put

$$F(s, A, H) = \sum_{\substack{d \bmod s \\ H \equiv b_* A d \pmod{s}}} \chi_s(d) g_{\xi\chi_A} \left(b_* A, \frac{H - b_* A d}{s} \right),$$

$$K_\eta(c, n, s, a; u) = \frac{F(s, cn^3, ua^6)\chi_s(b_*)}{\sqrt{qb_*}\tau\xi(s)\overline{\xi(u)}}\gamma_2(c)\overline{\alpha(cn^3)}\overline{G(cn^3)}\eta(cn^3)\Xi(cn^3)^{-1}R(cn^3, s),$$

$$\mathcal{F}_{\eta,u}(x, w, z) = \sum_{c \text{ sf}, n, s, a} K_\eta(c, n, s, a; u)q_c^{-1/2-x}q_n^{-1-3x}q_s^{-w}q_a^{-6z}.$$

To prove (5), use the full modulus sb_*A , even when s, A share primes. With the positive sign for the finite Fourier coefficient, the required transform is

$$\sum_{m \bmod sb_*A} \xi(m)\chi_A(m)q_s^{-1/2}g_{\chi_s}(s, -m)e\left(\frac{Hm}{sb_*A}\right) = q_s^{1/2}F(s, A, H).$$

To see this, write $m = r + b_*A\ell$, with $r \bmod b_*A$ and $\ell \bmod s$, and expand the s -Gauss sum. Additive orthogonality in ℓ imposes $H \equiv b_*Ad \pmod{s}$; the remaining sum in r is the Gauss sum in the definition of F . No coprimality between s and A is used. The planar dilation contributes X/q_A ; including the outer square root leaves $X^{1/2}/(\sqrt{qb_*}q_A)$. The radial Fourier argument is Xq_H/q_A . Primitivity of each local component of ξ removes all frequencies meeting S , including zero: at a prime in S , its Gauss frequency is a unit multiple of H . Write uniquely $H = ua^6$, with a primary and u retaining the unit of H ; thus this change of variables introduces no factor six. Mellin inversion and $x = t + 1 - z$ give (5); the initial lines $\operatorname{Re} t = 4$ and $\operatorname{Re} z = 2$ give $\operatorname{Re} x = 3$ without a contour shift. The trivial bound $|F| \leq q_sq_{b_*}q_A$ gives the summable majorant

$$C_S \sum_{u, c, n, s, a} q_u^{-2}q_c^{-5/2}q_n^{-7}q_s^{-2}q_a^{-12}$$

on $(\operatorname{Re} x, \operatorname{Re} w, \operatorname{Re} z) = (3, 3, 2)$. Together with the Gaussian and Mellin-weight decay, this justifies all sums and integrals used in the Poisson representation.

Fixed and moving phase cancellation. CRT gives

$$g_{\xi\chi_A}(b_*A, K) = \xi(A)\chi_A(b_*)g_\xi(b_*, K)g_{\chi_A}(A, K).$$

At S , $K = (H - b_*Ad)/s \equiv H/s$, so its factor is $\sqrt{qb_*}\tau\xi(A)\overline{\xi(u)}\xi(s)$. The denominator in K_η and $\Xi(A)^{-1}$ cancel all ξ factors. The remaining b_* symbols cancel as $\prod_p \chi_p(b_*)^{t_p - k_p} = \chi_A(b_*)/\chi_s(b_*)$. Here $t_p = v_p(A)$ and $k_p = v_p(s)$.

For the other CRT units write $h_p = H/p^{v_p(H)}$, $s_p = s/p^{k_p}$, and $A_p = A/p^{t_p}$. Their factor is

$$U_p = \chi_p(h_p)^{k_p - t_p}\chi_p(s_p)^{t_p}\chi_p(A_p)^{t_p - k_p}\chi_p(b_*)^{t_p - k_p}.$$

Applying (A.2) to distinct primes cancels the off-diagonal A, s phases exactly:

$$R(A, s) \prod_p \chi_p(s_p)^{t_p}\chi_p(A_p)^{-k_p} = \prod_p R(p, p)^{t_p k_p}.$$

Writing $t_p = e_p + 3l_p$ with $e_p = v_p(c)$, division by the pair factors of $\gamma_1(c)$ leaves a product over $p < r$, with factor $R(p, r)^{e_p l_r + e_r l_p + l_p l_r}$. The refinement law (A.1) expresses that product as

$$R(c, n)G(n^3) \prod_p \overline{G(p^3)}^{l_p} R(p, p)^{-e_p l_p - l_p(l_p - 1)/2}.$$

Its first two factors combine with $G(c)$ to cancel the inserted $\overline{G(cn^3)}$. Also $\alpha(c)\overline{\alpha(cn^3)} = \overline{\alpha(n)}^3$. These identities hold even when c, n share primes.

A.3 Local Euler factors

For $p \notin S$ set $Q = Np$, $j = v_p(u)$, $\rho = \chi_p(u/p^j)$, $\omega_p = \chi_p(-1)$, and

$$a_p = \overline{\alpha(p)}^3 \eta(p)^3, \quad b_p = \overline{\alpha(p)}^2 \eta(p)^2 \chi_p(4), \quad V = Q^{-6z}, \quad \mathcal{R} = a_p^2 Q^{4-6x-6z}, \quad W_\ell = \rho Q^{-w}.$$

Thus $b_p^3 = a_p^2$. The factor of $\mathcal{F}_{\eta,u}$ at p is

$$P_p^* = \frac{1}{1-\mathcal{R}} \left\{ \frac{\mathcal{R}(1-Q^{-1}) - \eta(p)(Q-1)Q^{-x-w}V^{\mathbf{1}_{j \leq 1}}}{1-V} + J_j \right\},$$

$$P_p = \frac{1}{1-V} + \mathbf{1}_{j=0} \frac{W_\ell}{1-W_\ell} + P_p^*,$$

where

j	J_j
0	$-\eta(p)\rho^{-1}Q^{-x} + W_\ell \mathcal{R}$
1	$\eta(p)Q^{-x-w}$
2	$a_p \rho^{-3} Q^{3/2-3x}$
3	$-\eta(p)b_p \rho^{-2} Q^{2-3x-w} + b_p^2 \rho^{-4} Q^{2-4x}$
4	$-\eta(p)a_p \rho^{-3} Q^{5/2-4x-w}$
5	$-a_p^2 Q^{3-6x}$.

(A.4)

Here is a direct finite verification of the table. For $t \geq 1$, summing over lifts of a residue modulo p gives, for any integer h ,

$$g_{\chi_p^h}(p^t, K) = \begin{cases} Q^{t-1} g_{\chi_p^h}(p, K/p^{t-1}), & p^{t-1} \mid K, \\ 0, & p^{t-1} \nmid K. \end{cases}$$

At prime modulus, a unit frequency κ contributes $\chi_p(\kappa)^{-h} g_{\chi_p^h}(p, 1)$; a zero frequency contributes 0 for $h \not\equiv 0 \pmod{6}$ and $Q-1$ otherwise. In the latter case the unit-frequency sum is -1 . Before restoring the units, the local convolution is

$$C_p(t, k, j') = \sum_{\substack{d \bmod p^k \\ p^{j'} \equiv p^t d \pmod{p^k}}} \chi_p(d)^k g_{\chi_p^t} \left(p^t, \frac{p^{j'} - p^t d}{p^k} \right).$$

When $k = 0$, the absent character factor is defined to be one; when $t = 0$, the modulus-one Gauss factor is one. These empty factors are distinct from zeroth powers at positive modulus, which retain their zero values at nonunits. It equals 1 for $t = k = 0$, $\mathbf{1}_{j'=0}$ for $t = 0 < k$, $g_{\chi_p^t}(p^t, p^{j'})$ for $k = 0 < t$, and $\omega_p^{-1} G_1 g_{\chi_p^{t-1}}(p^t, p^{j'-1})$ for $t > 0, k = 1, j' \geq 1$, where $G_1 = g_{\chi_p}(p, 1)$. For the last formula, expand the inner Gauss sum and sum over d : at a unit y modulo p the resulting factor is $\sum_{d \bmod p} \chi_p(d) e(-dy/p) = \omega_p^{-1} \chi_p(y)^{-1} G_1$. If $t > 0, k = 1, j' = 0$, the congruence has no solution. All other cases vanish. For $t > 0$ and $k \geq 2$, write $d = d_0 + p^{k-1}h$. The congruence and character are unchanged, while expanding the inner Gauss sum at a unit y produces $\sum_{h \bmod p} e(-hy/p) = 0$. The restored summand, with $t = e + 3l$, is

$$(-1)^e \gamma_1(p)^{-e} \eta(p)^e (a_p \overline{\gamma_3(p)})^l \rho^{k-t} \omega_p^{tk-el-l(l-1)/2} C_p(t, k, j+6m) Q^{-(x+1/2)e-(1+3x)l-wk-6zm}.$$

The possible positive exponents are $t = 6r + 6, 6r + 1, 6r + 3, 6r + 4$, with $r \geq 0$. The lifting formula gives the following complete list of frequencies for which $C_p(t, k, j')$ can be nonzero:

t	$k = 0$	$k = 1$
$6r + 6$	$j' \geq t - 1$	$j' = t$
$6r + 1$	$j' = t - 1$	$j' \geq t$
$6r + 3$	$j' = t - 1$	$j' = t$
$6r + 4$	$j' = t - 1$	$j' = t$

The equality at the lower endpoint uses a unit-frequency Gauss sum; the strict inequalities in the first two rows use the principal character's zero-frequency value $Q - 1$. Insert $j' = j + 6m$. Increasing r, m together by one multiplies a surviving summand by $\mathcal{R}$. The two strict-inequality tails give respectively

$$\frac{\mathcal{R}(1 - Q^{-1})}{(1 - \mathcal{R})(1 - V)}, \quad -\frac{\eta(p)(Q - 1)Q^{-x-w}V^{\mathbf{1}_{j \leq 1}}}{(1 - \mathcal{R})(1 - V)}.$$

For the second tail the first m is $r + 1$ if $j \leq 1$, and r otherwise; this explains the indicator in its power of V . The remaining equality cases, evaluated at $r = 0$ with the displayed restored summand, give $J_j/(1 - \mathcal{R})$ using the prime Gauss identities and $\gamma_2(p)\gamma_4(p) = 1$. For example, $j = 2$ has $t = 3, k = 0, m = 0$ and gives $a_p \rho^{-3} Q^{3/2-3x}$; $j = 3$ has the two cases $(t, k, m) = (3, 1, 0), (4, 0, 0)$, giving the two terms in J_3 . The $t = 0$ terms give the two geometric terms outside P_p^* . This proves the table for all exponents, including shared primes.

Finally, with the original zeros at $p \mid u$, put

$$W = \chi_p(u)Q^{-w}, \quad D = \eta(p)\overline{\chi_p(u)}Q^{-x}, \quad H_p = P_p \frac{(1 - V)(1 - W)}{1 - D}, \quad \mathcal{H}_{\eta, u} = \prod_{p \notin S} H_p.$$

Absolute convergence on the initial lines permits multiplication of the Euler factors. The factors $(1 - V)^{-1}$, $(1 - W)^{-1}$, and $1 - D$ give the zeta, numerator, and reciprocal factors in (6). At $p \mid u$, $W = D = 0$, retaining the prescribed deletions. This completes the proof of (5)–(6).

B The weak reflection formula

Proof of Lemma 3. Freeze P, s , the unit of m , and ν as in the lemma.

Theta coefficients and finite masks. Write $\mathbb{H}^3 = \{(z, v) : z \in \mathbb{C}, v > 0\}$, with its standard $\mathrm{SL}_2(\mathbb{C})$ action, and put $\check{e}(z) = \exp(2\pi i(z + \bar{z})) = e(\lambda z)$, $L(t) = \begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix}$, and $E = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. We use the cubic theta function of [2, (5.6)–(5.8)], normalized by

$$\theta(z, v) = \frac{3^{5/2}}{2} v^{2/3} + \sum_{\mu \neq 0} t_\theta(\mu) v K_{1/3}(4\pi|\mu|v) \check{e}(\mu z).$$

Here $K_{1/3}$ is the modified Bessel function of the second kind. Its coefficients are those of the cited formula (5.7); in particular, $t_\theta(\lambda^{-3}cb^3) = 3^{5/2}|b|\overline{\widetilde{g}_3(c)}$ for primary squarefree c and primary b . Here

$$\widetilde{g}_3(c) = q_c^{-1/2} \sum_{v \bmod c} (v/c)_3 \check{e}(v/c)$$

uses the ordinary cubic residue symbol, extended by zero on nonunits. It is defined even when c meets S , including the prime above 2, and has modulus one for squarefree c prime to λ , with

$\tilde{g}_3(1) = 1$. The infinity coefficients are even, so conjugation replaces $t_\theta(\mu)$ by $\overline{t_\theta(-\mu)} = \overline{t_\theta(\mu)}$. For each cusp representative H , define the dual coefficients by

$$\overline{\theta(H(z, v))} = \overline{\sigma_H} v^{2/3} + \sum_{\mu \neq 0} d_H(\mu) v K_{1/3}(4\pi|\mu|v) \check{e}(\mu z). \quad (\text{B.1})$$

For $H = I, L(\lambda), L(-\lambda)$ these are respectively the conjugate expansions with indices $j = 1, 19, 10$ in [2, (5.9), (5.12), (5.16) and Appendix A]. More precisely, $d_H(\mu) = \overline{d_j^{\text{DR}}(-\mu)}$. The coefficient $d(\mu)$ in each reflection branch is this $d_H(\mu)$.

Sextic reciprocity, including its zeros, writes the character of the completed index A as

$$\nu(A) \chi_A(m) = \Psi_0(A) \chi_R(A) \prod_{p|P, p \notin S} \chi_p(A)^{j_p}, \quad j_p \equiv v_p(P) \pmod{6}.$$

Here Ψ_0 ranges over a fixed finite family: it includes ν , the unit and S -prime symbols (with exponents modulo six), and the reciprocity factors involving P, R . A fixed ray subdivision of R fixes the latter. Choose a sufficiently divisible fixed modulus L , supported exactly on S and divisible by 18, for this family, the primary condition, and the cubic supplementary symbol. The function $\phi(x) = \chi_x(\lambda)^2 \Psi_0(x)$ on primary elements prime to S , zero elsewhere, is L -periodic. For $h_0 \in \mathcal{O}/L\mathcal{O}$, put

$$\widehat{\phi}(h_0) = q_L^{-1} \sum_{x \bmod L} \phi(x) e(-h_0 x/L), \quad \phi(x) = \sum_{h_0 \bmod L} \widehat{\phi}(h_0) e(h_0 x/L).$$

The pairing is nondegenerate and $|\widehat{\phi}(h_0)| \leq 1$. At the moving primes, finite Fourier inversion gives

$$C_{p,j}(h) = q_p^{-1} \sum_{x \bmod p} \chi_p(x)^j e(-hx/p) = \begin{cases} q_p^{-1/2} \tau_{p,j}^- \chi_p(h)^{-j}, & j \neq 0, h \neq 0, \\ 0, & j \neq 0, h = 0, \\ -q_p^{-1}, & j = 0, h \neq 0, \\ 1 - q_p^{-1}, & j = 0, h = 0, \end{cases}$$

where $\tau_{p,j}^- = q_p^{-1/2} \sum_{x \bmod p} \chi_p(x)^j e(-x/p)$ for $1 \leq j \leq 5$. These normalized Gauss factors have modulus one. Here j is represented by one of $0, \dots, 5$, and the power with $j = 0$ remains zero at $x = 0$. Thus even a principal local mask has a nonconstant Fourier expansion. Put $T = R \prod_{p|P, p \notin S} p$, and set $j_p = 1$ for $p \mid R$. The masked conjugate theta function is explicitly

$$\Theta_m(z, v) = \sum_{h_0 \bmod L} \sum_{(h_p \bmod p)_{p|T}} \widehat{\phi}(h_0) \prod_{p|T} C_{p,j_p}(h_p) \overline{\theta\left(z + \lambda^2 \left(h_0/L + \sum_{p|T} h_p/p\right), v\right)}.$$

Indeed, a translate by $\lambda^2 h/p$ multiplies the coefficient at μ by $e(\lambda^3 \mu h/p)$, as required. Group these terms according to the active frozen primes $p \mid D$. Each group consists of translates by

$$z_h = \lambda^2 \left(h_0/L + \sum_{p|DR} h_p/p \right), \quad h_p \neq 0.$$

All primes of R are active; a frozen prime can be inactive only if $j_p = 0$. For each h_0 there are at most $2^{\omega(P)}$ active sets; the h_p sums are evaluated as Gauss sums within each branch.

The primary mask on $x = \lambda^3 \mu$ retains precisely $\mu = \lambda^{-3} c b^3$: other ramified exponents have $\lambda \mid x$, and the negative branch has $x \equiv -1 \pmod{3}$. The retained coefficient is $3^{5/2} |b| \tilde{g}_3(c)$ with

$\tilde{g}_3(c) = \chi_c(\lambda)^{-2} \gamma_2(c)$. Since $\chi_b(\lambda)^6 = 1$, the supplementary factor in ϕ cancels this symbol. The whole-index mask excludes exactly the indices meeting S ; the moving characters act on cb^3 with their zeros.

By [2, (5.2)–(5.5)], the theta function satisfies $\theta(gw) = \kappa(g)\theta(w)$ on $\Gamma_2 = \langle \mathrm{SL}_2(\mathbb{Z}), \Gamma_1(3) \rangle$, where $\Gamma_1(3) = \{g \in \mathrm{SL}_2(\mathcal{O}) : g \equiv I \pmod{3}\}$ and $\kappa(g) = (c/a)_3$ on $\Gamma_1(3)$ for $c \neq 0$; it is one when $c = 0$ and on $\mathrm{SL}_2(\mathbb{Z})$.

Rational cusps and the multiplier. Reduce $\lambda^2 h_0/L = a_F/c_F$. Multiply this fixed pair by a unit so that $a_F \equiv 1 \pmod{3}$ if $\lambda \mid c_F$, and $c_F \equiv 1 \pmod{3}$ otherwise. This is possible because reduction identifies μ_6 with $(\mathcal{O}/3\mathcal{O})^\times$; the selected entry is a unit modulo 3. The choice depends only on h_0 . Put $r = DR$, and write

$$z_h = a/c, \quad c = c_F r, \quad a = a_F r + \lambda^2 c_F \sum_{p|r} (r/p) h_p.$$

The fraction is reduced: its numerator is $a_F r$ modulo primes of c_F and $\lambda^2 c_F (r/p) h_p$ modulo $p \mid r$, both units. Take $M = \lambda^{12} L^4$, lift $h_p \equiv 0 \pmod{M^2}$, and fix $R \pmod{M^2}$, hence $r = DR \pmod{M^2}$ since D is frozen. The lifts change z_h only by $\lambda^2 \mathcal{O} = 3\mathcal{O}$, a period with multiplier one. Complete a/c to $g = \begin{pmatrix} a & b_0 \\ c & \delta \end{pmatrix}$: choose $\delta \equiv a^{-1}$ at the prime-power parts of $M c_F$ supported on c_F , $\delta \equiv 0$ at the other prime-power parts of M , and $\delta \equiv a^{-1} \pmod{p}$ for $p \mid r$. Then $b_0 = (a\delta - 1)/c$, and all fixed-prime matrix residues are independent of the active h_p . Since r is primary and the moving terms contain λ^2 , a is primary when $\lambda \mid c$, and c is primary otherwise. The CRT choices then give $g \equiv I \pmod{3}$ if $\lambda^2 \mid c$, $g \equiv L(u_0) \pmod{3}$ with $u_0 = \pm \lambda \equiv c \pmod{3}$ if $v_\lambda(c) = 1$, and $g \equiv \begin{pmatrix} u_0 & -1 \\ 1 & 0 \end{pmatrix} \pmod{3}$ with $u_0 \equiv a \pmod{3}$ otherwise. Use these as H ; hence $gH^{-1} \in \Gamma_1(3)$. In the last case $H = EL(-u_0)$ and its theta function is one of the same three, since $E \in \mathrm{SL}_2(\mathbb{Z})$ and lower translations by $\mathbb{Z} + 3\mathcal{O}$ have multiplier one. All cusp coefficients and phases use this fixed normalized pair.

The published automorphy multiplier and cubic reciprocity give

$$\kappa(gH^{-1}) = \kappa_{\mathrm{fix}} \prod_{p|r} \chi_p(a)^2. \quad (\text{B.2})$$

Here κ_{fix} has modulus one and is constant on the fixed sector. The three theta expansions are those of $I, L(\lambda), L(-\lambda)$, since $\lambda - \omega^2, -\lambda - \omega \in \mathbb{Z} + 3\mathcal{O}$. For explicit verification, the fixed factors in the cases $v_\lambda(c) \geq 2$, $v_\lambda(c) = 1$, and $v_\lambda(c) = 0$ are respectively

$$(c_F/a)_3, \quad (-u_0/(a - u_0 b_0))_3 ((c_F/u_0)/a)_3, \quad (a/c_F)_3.$$

In the middle case $u_0 = \pm \lambda$ is congruent to c modulo 3; use $a(c - u_0 \delta) = c(a - u_0 b_0) - u_0$ and $b_0 c \equiv -1 \pmod{a}$. In detail, with $A_0 = a - u_0 b_0$ and $C_0 = c - u_0 \delta$, both a, A_0 are primary and

$$(C_0/A_0)_3 = (-u_0/A_0)_3 (a/A_0)_3^{-1} = (-u_0/A_0)_3 ((c/u_0)/a)_3.$$

Factoring $c/u_0 = (c_F/u_0)r$ and applying cubic reciprocity to a, r gives the stated fixed factor times $(a/r)_3$. In the last case the CRT choice gives $\delta \equiv 0 \pmod{9}$, hence $-b_0 c = 1 - a\delta \equiv 1 \pmod{9}$. Factor δ into its unit, λ -power, and primary part. Against the denominator $1 - a\delta$, the supplementary laws make the first two factors trivial; reciprocity makes the primary part trivial because $1 - a\delta \equiv 1 \pmod{\delta}$. Thus $(\delta/(-b_0 c))_3 = 1$, and

$$\kappa(gH^{-1}) = (\delta/(-b_0))_3 = (\delta/c)_3^{-1} = (a/c)_3 = (a/c_F)_3 (a/r)_3.$$

The chosen residues make each fixed factor independent of the active frequencies. Periodic changes of the cusp representative and of δ avoid zero matrix entries in these symbol formulas without changing any congruence.

The derivative and the reflected local characters. Write $k_g = gH^{-1} \in \Gamma_1(3)$. Since $g = k_g H$, $\overline{\theta(w)} = \overline{\kappa(k_g)} \overline{\theta(Hg^{-1}w)}$. For $w = (a/c + z, v)$ the coordinates of $g^{-1}w$ are

$$z' = -\frac{\delta}{c} - \frac{\bar{z}}{c^2(|z|^2 + v^2)}, \quad v' = \frac{v}{q_c(|z|^2 + v^2)}.$$

At $z = 0$, differentiation in $\bar{z}$ gives $\partial_{\bar{z}} z' = -(cv)^{-2}$ and $\partial_{\bar{z}} \bar{z}' = \partial_{\bar{z}} v' = 0$. It therefore becomes the z' derivative of the dual cusp series, at height $1/(q_c v)$, and kills every cusp constant: its derivative through v' is zero at $z = 0$. Termwise differentiation is valid by the locally uniform convergence of the exponentially decaying Bessel series for $v > 0$.

Put $x = \lambda^4 \mu$ and $f = \lambda^3 c_F$. Let δ_F be the residue of δ modulo f , and let r^{-1} denote the inverse of r modulo f . For $p \mid r$, put $\epsilon_p \equiv -(\lambda^5 c_F^2 (r/p)^2)^{-1} \pmod{p}$. Since $a \equiv \lambda^2 c_F (r/p) h_p \pmod{p}$ and $\delta \equiv a^{-1} \pmod{p}$, additive CRT gives

$$\check{e}(-\delta\mu/c) = \vartheta(x) \prod_{p \mid r} e(\epsilon_p h_p^{-1} x/p), \quad \vartheta(x) = e\left(-\frac{\delta_F r^{-1} x}{f}\right).$$

The residue δ_F and r^{-1} modulo f are fixed in the chosen sector, and $x \in \mathcal{O}$ on the entire dual support because $\mu \in \lambda^{-4} \mathcal{O}$. Thus this is a common column phase, including indices that meet S . Conjugating (B.2) contributes $\chi_p(h_p)^{-2}$. Combining with $C_{p,j}$ and putting $y = h_p^{-1}$ gives a Gauss sum of exponent $j + 2$. If this exponent is nonzero modulo six, the sum is zero at $p \mid x$ and is a unit-modulus scalar times $q_p^{1/2} \chi_p(x)^{-j-2}$ otherwise. If the exponent is zero, the sum is $-1 + q_p \mathbf{1}_{p \mid x}$. At a prime of R , $j = 1$, so its column factor is exactly $\chi_p(x)^{-3} = \chi_p(x)^3$. At a frozen active prime it is

$$B_p(x) = \begin{cases} \chi_p(x)^{-j_p-2}, & j_p \notin \{0, 4\}, \\ q_p^{-1/2}(-1 + q_p \mathbf{1}_{p \mid x}), & j_p = 4, \\ q_p^{-1/2} \chi_p(x)^{-2}, & j_p = 0. \end{cases}$$

The factors involving the units ϵ_p contribute only scalars of modulus one, absorbed into ζ_R ; in a Ramanujan sum they leave the condition $p \mid x$ unchanged. Thus the displayed B_p are independent of R , include all collisions, and satisfy the asserted bound.

Mellin normalization and convergence. Put $D_m(v) = \partial_{\bar{z}} \Theta_m(0, v)$, and let $\mathcal{D}_m(t)$ denote the series in (3) with $V(q_c q_b^3/Z)$ replaced by $(q_c q_b^3)^{-t}$. At $\mu = \lambda^{-3} c b^3$, $\bar{\mu} = -i |c b^3| \overline{\alpha(c b^3)} / (3\sqrt{3})$. The differentiated coefficient is consequently $6\pi |c b^3| |b| \gamma_2(c) \overline{\alpha(c b^3)}$ times the original character masks. Using $\int_0^\infty K_{1/3}(y) y^{w-1} dy = 2^{w-2} \Gamma((w-1/3)/2) \Gamma((w+1/3)/2)$ for $\operatorname{Re} w > 1/3$ gives

$$\int_0^\infty D_m(v) v^{2t} dv = \frac{81}{8\pi} \left(\frac{27}{4\pi^2} \right)^t \prod_{\pm} \Gamma(1+t \pm 1/6) \mathcal{D}_m(t) \quad (\text{B.3})$$

first for $\operatorname{Re} t > 1/2$, where both the coefficient series and the integrated Bessel series converge absolutely. In the reflected integral write $c_{\text{cusp}} = c_F D R$ to distinguish the cusp denominator from the squarefree summation variable c . The substitution $v' = 1/(q_{c_{\text{cusp}}} v)$ contributes $q_{c_{\text{cusp}}}^{1-2t}$, and the differentiated automorphy factor contributes $-c_{\text{cusp}}^{-2}$. The absolute conductor factor cancels because $c_{\text{cusp}}^{-2} q_{c_{\text{cusp}}} = \overline{\alpha(c_{\text{cusp}})}^2$. More explicitly, the moment of the derivative of a single translate, before inserting its finite Fourier weight, is

$$\begin{aligned} & \int_0^\infty \partial_{\bar{z}} \overline{\theta(a/c_{\text{cusp}} + z, v)} \Big|_{z=0} v^{2t} dv \\ &= -\frac{i}{8\pi} \overline{\kappa(k_g) \alpha(c_{\text{cusp}})}^2 (4\pi^2)^t q_{c_{\text{cusp}}}^{-2t} \prod_{\pm} \Gamma(1-t \pm 1/6) \sum_{\mu \neq 0} d_H(\mu) \alpha(\mu) q_\mu^{t-1/2} \check{e}(-\delta\mu/c_{\text{cusp}}), \end{aligned}$$

initially for $\operatorname{Re} t < -1/2$. Indeed, the published coefficient bound gives, for $\sigma > 1/2$,

$$\sum_{\mu \neq 0} |d_H(\mu)| q_\mu^{-\sigma-1/2} \ll \sum_{k \geq -4} 3^{-k(\sigma+1/3)} \sum_n q_n^{-\sigma-1/2} \sum_b q_b^{-3\sigma-1} < \infty.$$

The differentiated translates decay exponentially at both endpoints of the height integral, up to fixed powers of the height. At infinity this follows from the direct Bessel expansion; at zero it follows from the rational-cusp expansion after the derivative has removed the constant. Hence their Mellin moments are entire and bounded on each closed vertical strip. Dividing (B.3) by its gamma factors continues $\mathcal{D}_m(t)$ to an entire function, with strip bound $C_m(1 + |\Im t|)^A e^{\pi|\Im t|}$ by Stirling's formula. The constant may depend on the fixed row, which suffices for the identity. Multiplication by e^{t^2} makes the horizontal integrals tend to zero and justifies the following shift.

Divide by (B.3), shift the primal Mellin line to $\operatorname{Re} t = -\sigma$ with $\sigma > 1/2$, and replace t by $-t$. The dual series is absolutely convergent on that line, so it may be interchanged with the integral. This gives the kernel

$$V^\sharp(x) = \frac{1}{2\pi i} \int_{(\sigma)} e^{t^2} \prod_{\pm} \frac{\Gamma(1+t \pm 1/6)}{\Gamma(1-t \pm 1/6)} \left(\frac{(2\pi)^4 x}{27} \right)^{-t} dt.$$

The integral defining the kernel alone can then be moved to any fixed $\sigma \geq 0$; no interchange with a dual Dirichlet series on the line $\sigma = 0$ is required. The prefactor from the two Mellin normalizations is $-i\alpha(\overline{c_{\text{cusp}}})^2/81$; the remaining conductor power is entirely inside $V^\sharp(q_\mu Z/q_{\text{cusp}}^2)$. Every remaining scalar is a residue-symbol unit, a normalized Gauss sum, $\widehat{\phi}(h_0)$ of modulus at most one, or an inactive factor $1 - q_p^{-1}$. This proves $|\zeta_R| \leq 1/81$. The cusp formulas in [2, (5.7), (5.13), (5.14) and Appendix A] give the stated support. Their coefficient magnitudes, after removing $q_b^{1/2}$, are $3^{k/6+8/3}$ for $k = 3n - 4$ ($n \geq 1$), $3^{k/6+3}$ for $k = 3n - 3$ ($n \geq 0$), and 9 for $k = -4$ at the two other cusps. Each is at most $27 \cdot 3^{k/6}$, proving the uniform bound.

For the kernel, the numerator gamma factors have their first poles at $t = -5/6$ and $t = -7/6$, while reciprocal gamma factors are entire. There are therefore no poles in $\operatorname{Re} t \geq 0$. On $\operatorname{Re} t = \sigma \geq 0$, Stirling bounds the gamma quotient by $O_\sigma((1 + |\Im t|)^{4\sigma})$; the factor e^{t^2} supplies Gaussian decay. Euler differentiation only inserts powers of $-t$. The line $\sigma = 0$ gives boundedness, and the line $\sigma = A$ gives $O_{A,j}(x^{-A})$ for the j th Euler derivative. This proves the asserted uniform kernel bounds. The weighted dual sums in the reflection formula are absolutely convergent by these bounds and lattice counting. $\square$

Use of AI: The author’s main contributions are the following insights:

- Replace collision estimates with cubic theta reflection and quadratic Hecke mean-square bound.
- Replace the detector construction with fixed-height zero bins and Hecke zero-density theorem.
- Identify the principal row using CRT and conductor analysis. Avoid Chebotarev.
- Extract the principal signal through two successive residues. Control the remaining terms with uniform contour and tail estimates.

The author used Claude and ChatGPT to summarize and understand OpenAI’s proof and then prompted his insights to work out the details interactively and simplify the proofs one by one. Proofs generated by Claude were verified by Codex and vice-versa. The author verified the outputs from both and combined them into one unified proof using ChatGPT.

References

- [1] OpenAI, *The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane $\text{Re } s > 7/8$* , September 30, 2026, supplied manuscript. <https://github.com/openai/math/blob/main/preprints/The-Quasi-Riemann-Hypothesis-September-30-2026/paper.pdf>.
- [2] A. Dunn and M. Radziwiłł, *Bias in cubic Gauss sums: Patterson’s conjecture*, Annals of Mathematics 200 (2024), 967–1057. <https://arxiv.org/abs/2109.07463v3>.
- [3] L. Goldmakher and B. Louvel, *A quadratic large sieve inequality over number fields*, Mathematical Proceedings of the Cambridge Philosophical Society 154 (2013), no. 2, 193–212. <https://arxiv.org/abs/1112.1642v2>.
- [4] T. Khale, C. O’Kuhn, A. Panidapu, A. Sun, and S. Zhang, *A Bombieri–Vinogradov theorem for primes in short intervals and small sectors*, Journal of Number Theory 229 (2021), 142–167; <https://arxiv.org/abs/2008.09677v3>, Lemma 3.3.